

Policy optimization for CMDPs with bandit feedback: Best-of-both-worlds and beyond[★]

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ABSTRACT

We study online learning in *constrained Markov decision processes* (CMDPs) in which rewards and constraints may be either *stochastic* or *adversarial*. In such settings, Stradi et al. [1] proposed the first *best-of-both-worlds* algorithm able to seamlessly handle stochastic and adversarial constraints, achieving optimal regret and constraint violation bounds in both cases. This algorithm suffers from two major drawbacks. First, it only works under *full feedback*, which severely limits its applicability in practice. Moreover, it relies on optimizing over the space of occupancy measures, which requires solving convex optimization problems, an highly inefficient task. In this paper, we provide the first *best-of-both-worlds* algorithm for CMDPs with *bandit feedback*. Specifically, when the constraints are *stochastic*, the algorithm achieves $\tilde{O}(\sqrt{T})$ regret and constraint violation, while, when they are *adversarial*, it attains $\tilde{O}(\sqrt{T})$ constraint violation and a tight fraction of the optimal reward. Moreover, our algorithm is based on a policy optimization approach, which is much more efficient than occupancy-measure-based methods. Finally, we complement our analysis for the adversarial setting by showing the performance of our algorithm when the constraints are *non-stationary* (i.e., with bounded adversariality) and against a *weaker* baseline.

1. Introduction

Most of the learning tasks arising in real-world scenarios involve an agent sequentially interacting with an unknown environment. *Markov decision processes* (MDPs) [2] have emerged as the most natural models for such interactions, as they allow to capture the fundamental goal of learning an optimal (i.e., reward-maximizing) action-selection policy for the agent. However, in most of the real-world applications, the learner has to satisfy some additional requirements. For instance, in autonomous driving one has to avoid crashing with other cars [3], in ad auctions one must *not* deplete the allocated budget [4], while in recommendation systems offending items should *not* be presented to the users [5]. In order to capture such requirements, *constrained MDPs* (CMDPs) [6] have been introduced. These augment classical MDPs by adding costs that the agent is constrained to keep below some given thresholds.

[★] A short version of this article appeared in the 42nd International Conference on Machine Learning 2025 (ICML 2025). Specifically, in this version we introduce the novel analysis to handle the adversarial setting with bounded non-stationarity, which is provided in Section 4.2.1 and Appendix H. Moreover, in Appendix L, we provide the experimental evaluation of our algorithm.

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Over the last years, *online learning* problems in *episodic CMDPs* have received a growing attention (see, e.g., [7] for a seminal work in the field). These are problems in which the learner repeatedly interacts with the CMDP environment over multiple episodes. In such settings, the learner's goal is to minimize the *regret* of not always selecting a best-in-hindsight policy that satisfies cost constraints, while at the same time ensuring that the cumulative *violation* of cost constraints does *not* grow too fast over the episodes. Ideally, one would like that both the regret and the constraint violation grow sublinearly in the number of episodes T .

In online learning in episodic Markov decision processes, two different assumptions on how rewards and costs are determined at each episode are possible. They can be selected either *stochastically* according to fixed (unknown) probability distributions or *adversarially*, meaning that no statistical assumption is made.

Very recently, Stradi et al. [1] proposed the first *best-of-both-worlds* learning algorithm for online learning in episodic constrained MDPs. Such an algorithm is able to seamlessly handle stochastic and adversarial constraints, achieving optimal regret and violation bounds in both cases. However, it suffers from two major drawbacks. First, it only works under *full feedback*, meaning that the learning agent needs to observe rewards and costs defined over the whole environment after each episode. This is extremely unreasonable in practice, where only some feedback along the realized trajectory is usually available. Moreover, the algorithm works by optimizing over the space of occupancy measures, which requires solving a convex problem at every episode, an highly inefficient task.

1.1. On best-of-both-worlds guarantees

A distinctive feature of episodic CMDPs in online environments is that the nature of the costs can vary significantly across applications. In some scenarios, costs are well-modeled as *stochastic* (e.g., stationary dynamics), while in others they are better captured by an *adversarial* model (e.g., non-stationary or strategically perturbed constraints). Since the learner typically does *not* know *a priori* which regime is encountered, it is natural to seek *best-of-both-worlds* guarantees, namely, a *single* algorithm that is simultaneously optimal in the stochastic setting and robust in the adversarial one.

In the stochastic regime, the goal is to match the statistically optimal rates: one would like to achieve $\tilde{O}(\sqrt{T})$ regret and $\tilde{O}(\sqrt{T})$ cumulative constraint violation, both measured with respect to the optimal *feasible* benchmark. These rates are tight as shown in [8].

In the adversarial regime, instead, one cannot generally hope for the same notion of optimality while also enforcing sublinear violations as shown in [9]. Intuitively, an adversary can force the learner to “spend” episodes to maintain feasibility, inevitably reducing the attainable reward. For this reason, the appropriate target becomes a *competitive* guarantee: we aim at $\tilde{O}(\sqrt{T})$ violation while competing with the optimal *feasible* value up to a multiplicative factor depending on a Slater-type parameter ρ that quantifies the amount of strict feasibility, that is, ρ represents how much strictly feasible the most feasible solution is. Concretely, this leads to ensuring a constant fraction of the optimal feasible reward (specifically a $\rho/(1 + \rho)$ fraction), while maintaining $\tilde{O}(\sqrt{T})$ cumulative violation. Importantly, this trade-off is *tight*. Indeed, Bernasconi et al. [10] prove that, even in a two-action bandit instance with general long-term constraints, no algorithm can simultaneously guarantee sublinear cumulative violations and a competitive ratio strictly smaller than $1 + 1/\rho$ in the adversarial setting. Since such a bandit problem can be seen as a special case of episodic CMDPs (e.g., horizon $H = 1$), this lower bound directly applies to our setting, showing that achieving at most a $\rho/(1 + \rho)$ -fraction of the optimal reward (together with $o(T)$ violation) is unavoidable up to logarithmic factors.

1.2. Original contributions

We provide the first *best-of-both-worlds* algorithm for online learning in episodic CMDPs with *bandit feedback*. This means that, after each episode, the algorithm only needs to observe the realized rewards and costs along the trajectory traversed during that episode, as it is the case in most of the real-world applications. Moreover, our algorithm is based on a primal-dual policy optimization method, and, thus, it is arguably much more efficient than the one by Stradi et al. [1], as it does *not* require solving convex programs.

When the costs are *stochastic*, our algorithm attains $\tilde{O}(\sqrt{T})$ regret and constraint violation, while, when they are *adversarial*, it achieves $\tilde{O}(\sqrt{T})$ violation and a fraction of the optimal reward. These results match those of the full-feedback algorithm by Stradi et al. [1] and are provably tight. We also analyze the performances of our algorithm with respect to a parameter ρ measuring by “how much” Slater’s condition is satisfied. Specifically, if ρ is arbitrarily small, our algorithm can still guarantee $\tilde{O}(T^{3/4})$ regret and violation in the stochastic setting. Crucially, similarly to the algorithm by Stradi et al. [1], ours *does not require* any knowledge of the Slater’s parameter ρ . In order to obtain this result, we show that the Lagrangian multipliers are automatically bounded during the learning dynamics, by employing the *no-interval-regret* property of our primal and dual regret minimizers. Indeed, we develop the first algorithm for unconstrained MDPs with no-interval-regret, under *bandit feedback*. We believe that this result may also be of independent interest.

Differently from Stradi et al. [1], we also show that our algorithm attains $\tilde{O}(\sqrt{T} + B)$ regret and $\tilde{O}(\sqrt{T})$ violation in the adversarial setting with *limited adversariality*, where B is a measure of non-stationarity. In the same setting, when ρ is arbitrarily small, our algorithm attains $\tilde{O}(T^{1/4}(\sqrt{T} + B))$ regret and $\tilde{O}(T^{3/4})$ violation. Finally, we show that our algorithm can achieve sublinear regret and violation in the adversarial setting, when using a *weaker* baseline that has to satisfy the constraints at every round. Specifically, when ρ is large enough our algorithm attains $\tilde{O}(\sqrt{T})$ regret and violation, while it still achieves $\tilde{O}(T^{3/4})$ regret and violation when ρ is arbitrarily small.

1.3. Technical challenges

To achieve *best-of-both-worlds* guarantees in CMDPs with *bandit feedback* when the Slater’s parameter is unknown, this work faces several challenges, and it introduces novel approaches to overcome them. Specifically, the main technical challenges are highlighted in the following:

- **Bounding the Lagrange multipliers.** Bounding the Lagrange multipliers during the learning dynamics is necessary to guarantee that the primal regret minimizer has low regret, since any regret bound scales as the payoff range. This is done by employing no-interval regret algorithms for both primal and dual. Notice that the no-interval regret property is stronger than the classical no-regret one and requires a smoother learning dynamics. Guaranteeing the no-interval regret property in *bandit* settings harder than standard multi-armed bandits was an open problem in the literature.
- **Fixed share update in Markov decision processes.** To design a primal regret minimizer for MDPs with bandit feedback that enjoys the no-interval-regret property, we employ a policy-optimization approach based on online mirror descent (OMD) with an (unnormalized) negative-entropy regularizer, combined with a *fixed-share update* originally introduced by Cesa-Bianchi et al. [11]. On the one hand, OMD with negative entropy is a standard choice to handle bandit feedback. On the other hand, it is well known that it can incur *linear* interval regret when the learner’s strategies (in our case, policies) approach the boundary of the decision set (see Cesa-Bianchi et al. [11] for details). To address this issue, the fixed-share update replaces the pure OMD iterate with a convex combination of (i) the OMD solution to the per-episode optimization problem and (ii) a “uniform” reference point in the decision set. This has a stabilizing effect on the learning dynamics, as it prevents the learner from placing too much mass near the boundary. However, implementing fixed-share directly in the space of occupancy measures is problematic: the interpolation may fall outside the (estimated) occupancy-measure set, so projection-based techniques such as those in Stradi et al. [1] are *not* generally applicable in our setting. To overcome this obstacle, we design a *policy-based* primal algorithm, where the convex combination is always well defined. Employing fixed-share updates within policy-optimization methods for MDPs requires a novel analysis, which we report in full in [Appendix C](#).
- **Addressing the dependence on the primal loss range.** A key challenge in designing an effective primal algorithm is controlling how the primal regret bound depends on the range of the losses. Notice that this is not a trivial task, since the magnitude of the losses is unknown *a-priori*. Thus, by employing a standard approach, the regret would scale quadratically in the Lagrangian multipliers, as it depends on the square of the loss magnitude. To mitigate this and ensure the regret dependence on the Lagrangian multipliers is only linear, we introduce an *episode-dependent learning rate*. This parameter is dynamically adjusted to be proportional to the highest value of the Lagrangian multipliers observed up to the current episode.

1.4. Related works

In the following, we highlight the works that are mainly related to ours. We first provide some works in the field of unconstrained online MDPs (see [12–14] for some initial results on the topic). The setting studied in these works generally differentiates the problem based on the nature of the losses (either stochastic or adversarial), the knowledge of the transition probability, and the nature of the feedback. Usually two types of feedback are considered: in the *full-information feedback* model, the entire loss function is observed after the learner’s choice, while in the *bandit feedback* model, the learner only observes the loss due to the chosen action.

1.4.1. Online learning in MDPs

Azar et al. [15] study the problem of optimal exploration in episodic MDPs with unknown transitions, stochastic losses and bandit feedback. The authors improve the previous result by Auer et al. [12] designing an algorithm whose upper bound on the regret match the lower bound for this class, $\tilde{O}(\sqrt{T})$. Rosenberg and Mansour [16] studies the setting of episodic MDPs with adversarial losses, unknown transitions, and full information feedback. In this case the authors present an online algorithm exploiting entropic regularization and providing a regret upper bound of $\tilde{O}(\sqrt{T})$. The same setting is investigated when the feedback is bandit by Rosenberg and Mansour [17] who attain a regret upper bound of the order of $\tilde{O}(T^{3/4})$, which is improved by Jin et al. [18] by providing an algorithm that achieves in the same setting a regret upper bound of $\tilde{O}(\sqrt{T})$. Finally, Luo et al. [19] provide an optimal policy optimization algorithm for adversarial MDPs with bandit feedback.

1.4.2. Constrained online learning

In case of constrained problem, a fundamental result is presented by Mannor et al. [9], who show that it is impossible to attain both sublinear regret and constraints violations when both the losses and constraints are adversarial. To overcome such an impossibility result, Liakopoulos et al. [20] study a class of online learning problems with long-term budget constraints that can be chosen by an adversary and they define a new notion of regret. The new learner’s regret metric introduces the notion of a *K-benchmark*, *i.e.*, a comparator that meets the problem’s allotted budget over any window of length K . Castiglioni et al. [21,22] are the first to provide a *best-of-both-worlds* algorithm for online learning problems with long-term constraints (in single state settings), being the constraints stochastic or chosen by an adversary. Nonetheless, Castiglioni et al. [21] does not extend to general constraints, that is, it works for budget constraints only. Differently, Castiglioni et al. [22] study general constraints, but they assume the knowledge of the Slater’s parameter ρ , which is generally unreasonable in real-world settings.

1.4.3. Online learning in CMDPs

Constrained problems have been also studied in the context of CMDPs; however almost all previous works focus on the setting where the constraints are chosen stochastically. Wei et al. [23] study the case of episodic CMDPs with known transition probability, full-feedback, adversarial losses and stochastic constraints. The algorithm presented by the authors attains an upper bound both for constraints violation and for the regret of the order of $\tilde{O}(\sqrt{T})$. Zheng and Ratliff [24] present, in the setting of stochastic losses and constraints, where the transition probabilities are known and the feedback is bandit, an upper bound on the regret of their algorithm of the order of $\tilde{O}(T^{3/4})$, while the cumulative constraint violations is guaranteed to be below a threshold with a given probability. Bai et al. [25] provide the first algorithm to achieve sublinear regret when the transition probabilities are unknown, assuming that the rewards are deterministic and the constraints are stochastic with a particular structure. Efroni et al. [7] study the case where transition probabilities, rewards, and constraints are unknown and stochastic, while the feedback is bandit. The authors propose two approaches to deal with the exploration-exploitation dilemma in episodic CMDPs guaranteeing sublinear regret and constraint violations. Qiu et al. [26] provide a primal-dual approach based on *optimism in the face of uncertainty*. This work shows the effectiveness of such an approach when dealing with episodic CMDPs with adversarial losses and stochastic constraints, achieving both sublinear regret and constraint violation with full-information feedback. Liu et al. [27] study the case where the rewards and the constrained are stochastic with a sub-gaussian form, and it achieves regret $\tilde{O}(\sqrt{T})$ regret and zero violation when a strictly safe policy exists and it is known and a bounded violation when the strictly safe policy exists but it not known *a priori*. Ding et al. [28] design a primal-dual policy optimization no-regret algorithm for CMDPs with stochastic rewards and stochastic constraints. Wei et al. [29] design a model-free, simulator-free reinforcement learning algorithm for CMDPs that achieves regret of order $\tilde{O}(T^{4/5})$ with zero constraints violation, assuming the number of episodes to be exponentially large in $1/\rho$. Wei et al. [30], Ding and Laveai [31] and Stradi et al. [32] consider the case in which rewards and constraints are non-stationary, assuming that their variation is bounded. Thus, their results are *not* applicable to general adversarial settings. Ghosh et al. [33] attain sublinear regret and sublinear strong constraint violation, that is, not allowing for cancellations. Their algorithm works when both the rewards and the costs are stochastic; thus, these results are not applicable to our stochastic setting, as we allow the rewards to be adversarial. Similarly, Müller et al. [34] study the harder problem of bounding the regret and the constraints violation without allowing cancellation between episodes. They were the first to prove a sublinear regret guarantee without error cancellations for an efficient primal-dual algorithm in the online setup. Stradi et al. [35] attains optimal bounds in the same setting. However the whole analysis of both works is structured considering the feasibility parameter ρ as a constant. Stradi et al. [36], in the setting with adversarial losses, stochastic constraints and partial feedback, achieve sublinear regret and sublinear positive constraints violations. Recently, Stradi et al. [1] showed the first best-of-both-worlds (with respect to the constraints) algorithm for CMDPs. Precisely, the authors propose a primal-dual algorithm that optimizes over the occupancy measure space, under *full feedback*. When the constraints are stochastic, the algorithm achieves $\tilde{O}(\sqrt{T})$ regret and violation, both in the case in which rewards are adversarial and the one where they are stochastic. Contrariwise, in the adversarial setting, the algorithms attains $\tilde{O}(\sqrt{T})$ violations, and the no- α -regret property with $\alpha = \rho/(H+\rho)$, where ρ is a suitably-defined Slater's parameter.

Algorithm 1 Learner-Environment Interaction.

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1: for  $t = 1, \dots, T$  do
2:    $r_t$  and  $G_t$  are chosen stochastically or adversarially
3:   The learner chooses a policy  $\pi_t : X \times A \rightarrow [0, 1]$ 
4:   The state is initialized to  $x_0$ 
5:   for  $h = 0, \dots, H - 1$  do
6:     The learner plays  $a_h \sim \pi_t(\cdot | x_h)$ 
7:     The learner observes  $r_t(x_h, a_h)$  and  $g_{t,i}(x_h, a_h)$  for all  $i \in [m]$ 
8:     The environment evolves to  $x_{h+1} \sim P(\cdot | x_h, a_h)$ 
9:     The learner observes  $x_{h+1}$ 

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2. Problem setting

2.1. Online constrained Markov decision processes

An *online episodic constrained* MDPs (CMDPs) [6] is a tuple

$$M := (X, A, P, \{r_t\}_{t=1}^T, \{G_t\}_{t=1}^T).$$

Specifically, T is the number of episodes, with $t \in [T]$ denoting a specific episode. X, A are finite state and action spaces, respectively. $P : X \times A \rightarrow \Delta(X)$ is the transition function. We denote by $P(x' | x, a)$ the probability of going from state $x \in X$ to state $x' \in X$ by taking action $a \in A$. Notice that, w.l.o.g., in this work we consider *loop-free* CMDPs. Formally, this means that X is partitioned into H layers $X_0, \dots, X_H$ such that the first and the last layers are singletons, i.e., $X_0 = \{x_0\}$ and $X_H = \{x_H\}$, and that $P(x' | x, a) > 0$ only if $x' \in X_{h+1}$ and $x \in X_h$ for some $h \in [0 \dots H - 1]$. Any episodic CMDP with horizon H that is *not* loop-free can be cast into a loop-free one by suitably duplicating the state space H times. $\{r_t\}_{t=1}^T$ is a sequence of vectors describing the rewards at each episode $t \in [T]$, namely $r_t \in [0, 1]^{|X \times A|}$. We refer to the reward of a specific state-action pair $x \in X, a \in A$ for an episode $t \in [T]$ as $r_t(x, a)$. Rewards

may be either *stochastic*, in that case r_t is a random variable distributed according to a distribution $\mathcal{R}$ for every $t \in [T]$, or chosen by an *adversary*. $\{G_t\}_{t=1}^T$ is a sequence of constraint matrices describing the m *constraint* violations at each episode $t \in [T]$, namely $G_t \in [-1, 1]^{|X \times A| \times m}$, where non-positive violation values stand for satisfaction of the constraints. For $i \in [m]$, we refer to the violation of the i -th constraint for a specific state-action pair $x \in X, a \in A$ at episode $t \in [T]$ as $g_{t,i}(x, a)$. Constraint violations may be *stochastic*, in that case G_t is a random variable distributed according to a probability distribution $\mathcal{G}$ for every $t \in [T]$, or chosen by an *adversary*.

The learner chooses a *policy* $\pi : X \rightarrow \Delta(A)$ at each episode, defining a probability distribution over actions at each state. We denote by $\pi(\cdot|x)$ the probability distribution for a state $x \in X$, with $\pi(a|x)$ being the probability of action $a \in A$. In [Algorithm 1](#) we provide the interaction between the learner and the environment in a CMDP, where, for simplicity, we omit the dependence on t in the path traversed by the learner. Furthermore, we assume that the learner knows X and A , but they do *not* know anything about P . Notice that the interaction between the learner and the environment is with *bandit feedback*, namely, the rewards and the constraint violations are revealed for the traversed trajectory only.

Occupancy Measures. Given a transition function P and a policy π , the *occupancy measure* $q^{P,\pi} \in [0, 1]^{|X \times A \times X|}$ induced by P and π is such that, for all $x \in X_h, a \in A$, and $x' \in X_{h+1}$ with $h \in [0 \dots H - 1]$, it holds

$$q^{P,\pi}(x, a, x') = \mathbb{P}\{x_h = x, a_h = a, x_{h+1} = x' | P, \pi\}.$$

Moreover, we let

$$q^{P,\pi}(x, a) = \sum_{x' \in X_{h+1}} q^{P,\pi}(x, a, x'), q^{P,\pi}(x) = \sum_{a \in A} q^{P,\pi}(x, a).$$

Then, the set of *valid* occupancy measures can be characterized as follows [\[16\]](#). A vector $q \in [0, 1]^{|X \times A \times X|}$ is a valid occupancy measure of an episodic loop-free MDP if and only if the following three conditions hold:

$$\begin{aligned} (i) \quad & \sum_{x \in X_h} \sum_{a \in A} \sum_{x' \in X_{h+1}} q(x, a, x') = 1 \quad \forall h \in [0, \dots, H - 1] \\ (ii) \quad & \sum_{a \in A} \sum_{x' \in X_{h+1}} q(x, a, x') = \sum_{x' \in X_{h-1}} \sum_{a \in A} q(x', a, x) \\ & \quad \forall h \in [1, \dots, H - 1], \forall x \in X_h \\ (iii) \quad & P^q = P, \end{aligned}$$

where P is the transition function of the MDP and P^q is the one induced by q . Indeed, any valid occupancy measure q induces a transition function P^q and a policy π^q , defined as $P^q(x'|x, a) := \frac{q(x, a, x')}{q(x, a)}$ and $\pi^q(a|x) := \frac{q(x, a)}{q(x)}$.

2.2. Offline CMDPs baseline

In the following, we introduce the *offline* CMDP optimization problem, which is needed to define a proper baseline to evaluate the performances of online learning algorithms. Specifically, we introduce the following linear program parameterized by a reward vector r and a constraint matrix G :

$$\text{OPT}_{r,G} := \begin{cases} \max_{q \in \Delta(M)} & r^\top q \\ \text{s.t.} & G^\top q \leq \underline{0}, \end{cases} \quad (1)$$

where $q \in [0, 1]^{|X \times A|}$ is an occupancy measure and $\Delta(M)$ is the set of valid occupancy measures.

Furthermore, we state the following well-known condition on the offline CMDP problem.

Condition 1 (Slater's condition). *Given a constraint matrix G , the Slater's condition holds when there is a strictly feasible solution q° such that $G^\top q^\circ < \underline{0}$.*

Notice that, in this work, we do *not* assume that the Slater's condition holds. Indeed, our algorithm still works when a strictly feasible solution does *not* exist. We refer to [Section 2.4](#) for further details on this. Finally, we define the Lagrangian function of Problem (1), as follows.

Definition 1 (Lagrangian function). Given a reward vector r and a constraint matrix G , the Lagrangian function $\mathcal{L}_{r,G} : \Delta(M) \times \mathbb{R}_{\geq 0}^m \rightarrow \mathbb{R}$ of Problem (1) is defined as:

$$\mathcal{L}_{r,G}(q, \lambda) := r^\top q - \lambda^\top (G^\top q).$$

2.3. Online learning problem

As it is standard in online learning [\[37\]](#), we evaluate the performance of learning algorithms by means of the notion of cumulative regret.

Definition 2 (Regret). We define the cumulative regret up to episode T as follows:

$$R_T := T \text{OPT}_{\bar{r}, \bar{G}} - \sum_{t=1}^T r_t^\top q^{P, \pi_t},$$

where $\bar{r} := \mathbb{E}_{r \sim \mathcal{R}}[r]$ if the rewards are stochastic, while $\bar{r} := \frac{1}{T} \sum_{t=1}^T r_t$ if they are adversarial, and $\bar{G} := \mathbb{E}_{G \sim \mathcal{G}}[G]$ if the constraints are stochastic, while $\bar{G} := \frac{1}{T} \sum_{t=1}^T G_t$ if they are adversarial.

We refer to an optimal occupancy measure, i.e., a feasible one achieving value $\text{OPT}_{\bar{r}, \bar{G}}$, as q^* . Thus, we can rewrite the regret definition as $R_T = \sum_{t=1}^T \bar{r}^\top q^* - \sum_{t=1}^T r_t^\top q^{P, \pi_t}$. Notice that, in the adversarial setting, the regret is computed with respect to an *optimal feasible strategy in hindsight*. Indeed, an optimal solution is *not* required to satisfy the constraints at every episode $t \in [T]$.

Next, we define the performance measure related to constraints: the *cumulative constraint violation*.

Definition 3 (Constraint violation). The cumulative constraint violation up to episode T is defined as:

$$V_T := \max_{i \in [m]} \sum_{t=1}^T [G_t^\top q^{P, \pi_t}]_i.$$

Learning algorithms perform properly when they are capable of keeping both the quantities defined above sublinear in T , namely, $R_T = o(T)$ and $V_T = o(T)$.

For the sake of simplicity, in the rest of the paper, we will refer to q^{P, π_t} as q_t , omitting the dependence on transition function P and policy π .

2.4. Feasibility

We introduce a problem-specific parameter of Problem (1), called $\rho \in [0, H]$, which identifies by “how much” Slater’s condition is satisfied. Formally:

- When the constraints are selected *stochastically*, namely, they are chosen from a fixed distribution, the parameter ρ is defined as

$$\rho := \max_{q \in \Delta(M)} \min_{i \in [m]} -[\bar{G}^\top q]_i.$$

- When the constraints are chosen *adversarially*, namely, no statistical assumption is made, the parameter ρ is defined as

$$\rho := \max_{q \in \Delta(M)} \min_{i \in [T]} \min_{i \in [m]} -[G_i^\top q]_i.$$

Furthermore, we denote the occupancy measure $q \in \Delta(M)$ leading to the value of ρ by q° . Intuitively, ρ represents the “margin” by which the “most feasible” strictly feasible solution (i.e., q°) satisfies the constraints. Finally, we state the following condition on the parameter ρ , which will guide the analyses of the performances of our algorithm.

Condition 2. It holds that $\rho \geq T^{-\frac{1}{8}} H \sqrt{112m}$.

Remark 1. Notice that, it is standard in the literature of primal-dual methods to assume that ρ is a constant independent of T and directly include in the regret bound the dependence on $1/\rho$ (see, e.g., [7,27,29,34]). Nevertheless, when ρ is too small, this could result in suboptimal regret bounds. In this paper, we take a different approach by providing bounds for *any value of ρ* , that is, whether Condition 2 holds or not.

3. A policy optimization primal-dual approach

In this section, we provide the description of our algorithm. We resort to a primal-dual formulation of the CMDP problem, and we employ different regret minimizers to optimize over the primal space (namely, the policy space) and the dual one (that is, the Lagrangian variables space). Furthermore, our primal algorithm is based on a policy optimization approach. Thus, the learning update is *not* performed over the occupancy measure space, but state-by-state along the MDP structure. This allows us to avoid solving a convex program at each episode (as it is the case in the algorithm by [1]). As concerns the dual, we employ online gradient descent (OGD). We remark that our algorithm does *not* require any knowledge of the Slater’s parameter ρ . Indeed, as we further discuss in the rest of this work, we can show that the Lagrangian multipliers are automatically bounded given specific no-regret properties of the primal and dual regret minimizers.

3.1. Meta-algorithm

In Algorithm 2, we provide the pseudocode of *primal-dual bandit policy search* (PDB-PS).

Algorithm 2 initializes the policy uniformly over the space (see Line 1). Moreover, the Lagrangian variables are initialized as the zero vector, the loss scaling factor to 1, the loss range to 2, and, finally, the dual space is instantiated as $[0, T^{1/4}]^m$ (see Line 3). We underline that we force the dual space to be bounded in $[0, T^{1/4}]^m$ only to deal with degenerate cases where Condition 2 does *not*

Algorithm 2 PDB-PS.

Require: State space X , action space A , number of episodes T , confidence parameter $\delta \in (0, 1)$

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1:  $\pi_1(a|x) \leftarrow \frac{1}{|A|} \quad \forall (x, a) \in X \times A$ 
2:  $\lambda_1 \leftarrow 0, \Gamma_1 \leftarrow 1, \Xi_1 \leftarrow 2$ 
3:  $\mathcal{K} \leftarrow [0, T^{1/4}]^m, \eta \leftarrow \frac{1}{D \ln(|A||X|^2 T^2 / \delta) \sqrt{T}}$ 
4: for  $t = 1, \dots, T$  do
5:   Play policy  $\pi_t$ , observe trajectory  $\{(x_h, a_h)\}_{h=0}^{H-1}$ , rewards  $\{r_t(x_h, a_h)\}_{h=0}^{H-1}$  and constraint violations  $\{g_{t,i}(x_h, a_h)\}_{h=0}^{H-1}$  for all  $i \in [m]$ 

6:   for  $h = 0, \dots, H-1$  do
7:      $\ell_t(x_h, a_h) \leftarrow \Gamma_t + \sum_{i=1}^m \lambda_{t,i} g_{t,i}(x_h, a_h) - r_t(x_h, a_h)$ 
8:      $\pi_{t+1} \leftarrow \text{Call FS-PODB.UPDATE}(\{(x_h, a_h)\}_{h=0}^{H-1}, \{\ell_t(x_h, a_h)\}_{h=0}^{H-1}, \Xi_t)$ 
9:      $\lambda_{t+1} \leftarrow \Pi_{\mathcal{K}} \left[ \lambda_t + \eta \sum_{h=0}^{H-1} G_t[x_h, a_h] \right]$ 
10:     $\Gamma_{t+1} \leftarrow 1 + \|\lambda_{t+1}\|_1$ 
11:     $\Xi_{t+1} \leftarrow \max \{\Xi_t, 2\Gamma_t\}$ 

```

hold. When Condition 2 holds, our algorithm guarantees that the Lagrangian variables are automatically bounded during learning. Furthermore, the algorithm keeps track of the maximum loss range observed by the primal algorithm Ξ_t , up to episode $t \in [T]$, since the primal regret minimizer needs to dynamically update its belief on the loss range, in order to attain optimal regret bounds. The algorithm plays policy π_t and observes the *bandit* feedback as depicted in [Algorithm 1](#) (see Line 5). Given the observed feedback, PDB-PS builds a re-scaled Lagrangian loss for each layer $h \in [H]$ as:

$$\ell_t(x_h, a_h) := \Gamma_t + \sum_{i=1}^m \lambda_{t,i} g_{t,i}(x_h, a_h) - r_t(x_h, a_h). \quad (2)$$

Notice that the loss built in [Eq. \(2\)](#) can be seen as the Lagrangian suffered by π_t for state-action pair (x, a) , scaled by Γ_t to guarantee that the losses are always positive (see Line 7). This loss is properly built to feed the primal policy optimization procedure. Moreover, we underline that the feedback given to the primal algorithm encompasses the trajectory and the maximum loss range observed, besides the loss built in [Eq. \(2\)](#). Policy π_{t+1} is returned by the primal algorithm (Line 8). We refer the reader to the next section for further discussion on the primal optimization algorithm. [Algorithm 2](#) updates the Lagrangian multipliers using an online gradient descent update with loss $-\sum_{h=0}^H G_t[x_h, a_h]$ in the bounded dual space $[0, T^{1/4}]^m$:

$$\lambda_{t+1} \leftarrow \Pi_{\mathcal{K}} \left[\lambda_t + \eta \sum_{h=0}^{H-1} G_t[x_h, a_h] \right],$$

where $\Pi_{\mathcal{K}}$ is the euclidean projection over the space $\mathcal{K}$ and $G_t[x_h, a_h]$ is the m -dimensional vector composed by the violations of any constraint for the state-action pair (x_h, a_h) (Line 9). Thus, the current loss scaling factor is computed as $\Gamma_{t+1} \leftarrow 1 + \|\lambda_{t+1}\|_1$ (Line 10). Finally, the maximum observed loss range Ξ_{t+1} is updated as $\Xi_{t+1} \leftarrow \max \{\Xi_t, 2\Gamma_{t+1}\}$, since the range of losses observed by the primal depends on the Lagrangian multipliers (Line 11).

3.2. Primal regret minimizer

In [Algorithm 3](#), we provide the pseudocode of *fixed share policy optimization with dilated bonus* (FS-PODB.UPDATE), namely, the update performed by the primal regret minimizer employed by [Algorithm 2](#). [Algorithm 3](#) builds on top of the state-of-the-art policy optimization algorithm for adversarial MDPs (see [\[19\]](#)), equipping it with a fixed share update [\[11\]](#). This modification allows us to achieve the no-interval regret property, which, to the best of our knowledge, has never been shown for adversarial MDPs with bandit feedback. Thus, we believe that the theoretical guarantees of [Algorithm 3](#) are of independent interest.

Specifically, [Algorithm 3](#) requires in input the trajectory traversed during the learner-environment interaction, the incurred loss functions, and the maximum loss range observed for any $t \in [T]$.¹ During the first episode, the algorithm initializes the estimated transitions space as the set of all possible transition functions (Line 2). Thus, at each episode the algorithm defines a dynamic learning rate $\eta_t \propto \frac{1}{\sqrt{T\Xi_t}}$ (Line 3), where Ξ_t is the upper bound on the range of the loss functions up to t . This is done to control the different scales of the loss, due to the Lagrangian multipliers choice of the dual algorithm. Then, [Algorithm 3](#) builds an *optimistic* estimator of the state-action value function as:

$$\hat{Q}_t(x, a) := \frac{L_{t,h}}{q_t(x, a) + \gamma} \parallel_t(x, a),$$

¹ While the input of [Algorithm 3](#) may seem different from the standard bandit feedback received in adversarial MDPs, this is *not* the case. Indeed, it is sufficient to set $\Xi_t = 1$ for all $t \in [T]$ to achieve the same guarantees attained by [Algorithm 3](#), in the Lagrangian formulation of CMDPs, in standard adversarial MDPs.

where $\mathbb{I}_t(x, a) := \mathbb{I}\{x_{t,h} = x, a_{t,h} = a\}$ and $L_{t,h} := \sum_{j=h}^{H-1} \ell_t(x_j, a_j)$ is the loss incurred by the algorithm at episode t starting from layer h . Indeed, since $\bar{q}_t(x, a) := \max_{\hat{P}_t} q^{\hat{P}_t, \pi_t}(x, a)$,² and γ is a positive quantity, $\hat{Q}_t(x, a)$ results in an optimistic estimator of the state-action

Algorithm 3 FS-PODB.UPDATE.

Require: Observed trajectory $\{(x_h, a_h)\}_{h=0}^{H-1}$, observed losses $\{\ell_t(x_h, a_h)\}_{h=0}^{H-1}$, loss range upper bound Ξ_t

1: **if** $t = 1$ **then**

2: $\mathcal{P}_1 \leftarrow$ set of all possible transitions

3: $\eta_t \leftarrow \frac{1}{2H\Xi_t C\sqrt{T}}$, $\gamma \leftarrow \frac{1}{C\sqrt{T}}$, $\sigma \leftarrow \frac{1}{T}$

4: For all $h = 0, \dots, H-1$ and $(x, a) \in X_h \times A$:

$$L_{t,h} \leftarrow \sum_{j=h}^{H-1} \ell_t(x_j, a_j)$$

$$\hat{Q}_t(x, a) \leftarrow \frac{L_{t,h}}{\bar{q}_t(x, a) + \gamma} \mathbb{I}_t(x, a),$$

where we let $\bar{q}_t(x, a) := \max_{\hat{P}_t} q^{\hat{P}_t, \pi_t}(x, a)$ and $\mathbb{I}_t(x, a) := \mathbb{I}\{x_{t,h} = x, a_{t,h} = a\}$

5: For all $(x, a) \in X \times A$:

$$b_t(x) \leftarrow \mathbb{E}_{a \sim \pi_t(\cdot|x)} \left[\frac{3\gamma H \Xi_t + H \Xi_t (\bar{q}_t(x, a) - \underline{q}_t(x, a))}{\bar{q}_t(x, a) + \gamma} \right]$$

$$B_t(x, a) \leftarrow b_t(x) +$$

$$\left(1 + \frac{1}{H}\right) \max_{\hat{P}_t} \mathbb{E}_{x' \sim \hat{P}_t(\cdot|x, a)} \mathbb{E}_{a' \sim \pi_t(\cdot|x')} [B_t(x', a')]$$

where we let $\underline{q}_t(x, a) := \min_{\hat{P}_t} q^{\hat{P}_t, \pi_t}(x, a)$, and $B_t(x_H, a) := 0$ for all $a \in A$

6: For all $(x, a) \in X \times A$:

$$\begin{aligned} w_{t+1}(a|x) &\leftarrow (1 - \sigma) w_t(a|x) e^{-\eta_t (\hat{Q}_t(x, a) - B_t(x, a))} \\ &\quad + \frac{\sigma}{|A|} \sum_{a' \in A} w_t(a'|x) e^{-\eta_t (\hat{Q}_t(x, a') - B_t(x, a'))} \\ \pi_{t+1}(a|x) &\leftarrow \frac{w_{t+1}(a|x)}{\sum_{a' \in A} w_{t+1}(a'|x)} \end{aligned}$$

7: $\mathcal{P}_{t+1} \leftarrow \text{TRANSITION.UPDATE}(\{(x_h, a_h)\}_{h=0}^{H-1})$

value function (Line 4). The optimistic estimator is employed to control the variance of the loss estimation and, thus, in order to achieve high-probability results. Finally, notice that the state-action value function (as the estimated one) is commonly used in policy optimization as it allows to optimize efficiently state-by-state. In addition to the estimated state-action value function, [Algorithm 3](#) defines a dilated bonus similar to the one introduced by Luo et al. [19], which is then incorporated in the final objective of the optimization update. The bonus is defined as:

$$B_t(x, a) := b_t(x) + \left(1 + \frac{1}{H}\right) \max_{\hat{P}_t} \mathbb{E}_{x' \sim \hat{P}_t(\cdot|x, a)} \mathbb{E}_{a' \sim \pi_t(\cdot|x')} [B_t(x', a')],$$

where the term $b_t(x)$ depends on the uncertainty on the transitions estimation and the range of the losses, while the term $\left(1 + \frac{1}{H}\right)$ attributes more weight to the deeper layers, so as to incentivize exploration (Line 5). The weights associated to any action are computed employing the so called fixed share update [11]; specifically, the weights are computed as the convex combination between the uniform weight and the solution to optimization step $\propto w_t(a|x) e^{-\eta_t (\hat{Q}_t(x, a) - B_t(x, a))}$. The policy is simply computed as a normalization between weights (see Line 6). Notice that the convex combination mentioned above is crucial to bound the regret for each interval (that is, to attain the no-interval regret property). Indeed, it guarantees a lower bound for the value taken by the policy in each available action at each episode, and, thus, for all intervals $[t_1, t_2] \subset [T]$, it allows to find a nice upper bound for the Bregman divergence $D_\psi(\pi(\cdot|a); \pi_{t_1}(\cdot|a))$, for all policies π . Finally, the estimation of the transitions is updated given the trajectory traversed in the MDP (Line 7). This estimation is standard in the literature. Thus, we refer to [16] for further discussion on the use of counters and epochs to estimate a superset of the transition space $\mathcal{P}_t$.

² As shown in [18], $\bar{q}_t(x, a)$ can be computed efficiently by means of dynamic programming.

3.3. No-interval regret property

When the Slater's parameter ρ is known, the only necessary requirement for the primal and the dual regret minimizers is to be no-regret. Thus, it is sufficient to bound the Lagrangian space so that $\|\lambda\|_1 \leq \mathcal{O}(H/\rho)$ to attain sublinear regret and violation. Nevertheless, knowing ρ is generally *not* possible in real-world scenarios. In order to relax the assumption on the knowledge of ρ , we require our primal and dual regret minimizers to have the no-interval regret property.³

First, we introduce the interval regret as follows.

Definition 4 (Interval regret). Given an interval of consecutive episodes $[t_1, \dots, t_2] \subseteq [1, \dots, T]$, the *interval regret* with respect to a general occupancy q (and the associated policy π) and a sequence of loss functions $\{\ell_t\}_{t=1}^T$ with $\ell_t : X \times A \rightarrow [0, K]$, with $K > 0$, is

$$R_{t_1, t_2}(q) := \sum_{t=t_1}^{t_2} \ell_t^\top(q_t - q).$$

In the following, we omit the dependence on the general occupancy measure q when it is clear from the context. Thus, given [Definition 4](#), we are able to introduce the no-interval regret property, as follows.

Definition 5 (No-interval regret property). An algorithm attains the no-interval regret property when for any interval of consecutive episodes $[t_1, \dots, t_2] \subseteq [1, \dots, T]$ and with respect to any valid occupancy q (and the associated policy π), it holds

$$R_{t_1, t_2} \leq \tilde{\mathcal{O}}(\sqrt{T}).$$

Intuitively, the no-interval regret property guarantees a more stable learning dynamics over the episodes.

When *full feedback* is available, as for the dual algorithm, it is sufficient to employ OGD-like updates to attain the desired result. This is *not* the case when the feedback is *bandit*. Nevertheless, given that we use a policy optimization procedure and the fixed share update, we build the first algorithm for adversarial MDPs with no-interval-regret. We state the result in the following theorem.

Theorem 1. For any $\delta \in (0, 1)$, with probability at least $1 - 8\delta$, Algorithm FS-PODB attains:

$$R_{t_1, t_2} \leq \tilde{\mathcal{O}}\left(\Xi_{t_1, t_2} \sqrt{T} + \Xi_{t_1, t_2} \frac{t_2 - t_1}{\sqrt{T}}\right),$$

where the regret can be computed with respect to any policy function $\pi : X \rightarrow \Delta(A)$.

As it is standard for online learning algorithms, R_{t_1, t_2} scales as the loss range, as shown by the dependence on Ξ_{t_1, t_2} , that is, the maximum possible range of losses in the interval.

3.4. Bound on the Lagrangian multipliers dynamics

Next, we show that, given the no-interval regret property of the primal and the dual regret minimizers, it is possible to show that the Lagrangian multipliers are automatically bounded during learning. Notice that this is necessary since any adversarial regret minimizer needs the loss to be bounded to achieve the no-regret property. Thus, since the rewards $\{r_t\}_{t=1}^T$ and the constraints $\{G_t\}_{t=1}^T$ are assumed to be bounded for all episodes, the problem of bounding the loss suffered by the primal algorithm becomes the problem of bounding the Lagrangian multipliers $\{\lambda_t\}_{t=1}^T$.

Theorem 2. Under [Condition 2](#), for any $\delta \in (0, 1)$, with probability at least $1 - 11\delta$, it holds:

$$\|\lambda_t\|_1 \leq \Lambda \quad \forall t \in [T + 1],$$

where $\Lambda = \frac{112mH^2}{\rho^2}$.

The general idea behind the proof is to compare, for every interval $[t_1, t_2] \subset [T]$, the upper bound to $-\sum_{t=t_1}^{t_2} \ell_t^{\mathcal{L}, \top} q_t$ obtained through the regret of the dual algorithm with the lower bound to the same quantity obtained through the primal interval regret, where we define the non-scaled Lagrangian loss $\ell_t^{\mathcal{L}}$ as the vector composed by

$$\ell_t^{\mathcal{L}}(x, a) := \sum_{i=1}^m \lambda_{t,i} g_{t,i}(x, a) - r_t(x, a)$$

for all $(x, a) \in X \times A$ and for all $t \in [T]$. The resulting inequality leads, by contradiction, to the desired bound. In this sense, a fundamental requirement for the proof is that the regret guarantees for both the primal and the dual algorithm hold for all subsets of episodes.

³ What we require is generally known in the literature as the *weak* no-interval regret property. For the sake of simplicity, in our work, we introduce only the *weak* property.

4. Theoretical analysis

In this section, we prove the best-of-both-world guarantees attained by [Algorithm 2](#). In the main paper, we state the bounds highlighting their dependence on T and on the maximum Lagrange multiplier Λ . We defer the complete expressions to the appendix, and we refer to [Appendix B](#) for a dictionary of the constants.

4.1. Stochastic setting

We first study the performance of [Algorithm 2](#) when the constraints are stochastic.

In such a setting, our algorithm can handle two scenarios. In both of them, employing a primal-dual analysis shows that both the regret and the violations are bounded with order $\tilde{\mathcal{O}}(\sqrt{T})$ times the maximum value taken over all episodes of the Lagrangian multipliers, i.e. $\max_{t \in [T]} \|\lambda_t\|_1$. In the first scenario, [Condition 2](#) holds and thus we can apply [Theorem 2](#) to show that the Lagrangian multipliers are bounded. In such a case, $\max_{t \in [T]} \|\lambda_t\|_1$ can be easily bounded by Λ . When [Conditions 2](#) does not hold, we need to resort to the bound of Lagrangian multipliers derived by the instantiation of OGD decision space, leading to $\tilde{\mathcal{O}}(T^{3/4})$ regret and violations bounds.

Specifically, when [Condition 2](#) holds, the Lagrangian multipliers are nicely bounded by Λ .

Theorem 3. Suppose that [Condition 2](#) holds and the constraints are generated stochastically. Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{\mathcal{O}}(\Lambda\sqrt{T}), \quad V_T \leq \tilde{\mathcal{O}}(\Lambda\sqrt{T}),$$

with probability at least $1 - 14\delta$ when the rewards are stochastic, and with probability at least $1 - 13\delta$ when the rewards are adversarial.

When [Condition 2](#) does not hold, we can still use the bound forced by [Algorithm 2](#) on the dual space. Therefore, the Lagrangian multipliers are bounded by $mT^{1/4}$, leading to the following result.

Theorem 4. Suppose that [Condition 2](#) does not hold and the constraints are generated stochastically. Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{\mathcal{O}}(T^{3/4}), \quad V_T \leq \tilde{\mathcal{O}}(T^{3/4}),$$

with probability at least $1 - 11\delta$ when the rewards are stochastic, and with probability at least $1 - 10\delta$ when the rewards are adversarial.

4.2. Adversarial setting

We then study the performance of [Algorithm 2](#) when the constraints are adversarial.

Notice that, in such a setting, there exists an impossibility result from [9] that prevents any algorithm from attaining both sublinear regret and sublinear violations. Thus, best-of-both-worlds algorithms in constrained settings focus on attaining sublinear violations and a fraction of the optimal rewards (see e.g., [1,22]).⁴

In such a setting, we can show the following result.

Theorem 5. Suppose [Condition 2](#) holds and the constraints are adversarial. Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$\sum_{t=1}^T r_t^\top q_t \geq \Omega\left(\frac{\rho}{\rho + H} \cdot \text{OPT}_{\bar{r}, \bar{G}}\right), \quad V_T \leq \tilde{\mathcal{O}}(\Lambda\sqrt{T}),$$

with probability at least $1 - 14\delta$ when the rewards are stochastic, and with probability at least $1 - 13\delta$ when the rewards are adversarial.

4.2.1. Limited adversariality

In this section, we present the theoretical guarantees of [Algorithm 2](#) when the adversariality of the constraints is parameterized given a measure of adversariality, i.e., the distance between the actual constraints selected by the adversary and the closest fixed constraints matrix. This is defined as follows:

Definition 6. We define a measure of adversariality as:

$$B := \min_{G \in [-1, 1]^{|X \times A| \times m}} \sum_{t=1}^T \|G_t - G\|_1$$

We first show the performance of our algorithm when [Condition 2](#) holds.

Theorem 6. Suppose that [Condition 2](#) holds and the constraints are generated adversarially and are parameterized given B . Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains the following bounds:

$$R_T \leq \tilde{\mathcal{O}}(\Lambda(\sqrt{T} + B)), \quad V_T \leq \tilde{\mathcal{O}}(\Lambda\sqrt{T}),$$

with probability at least $1 - 14\delta$ when the rewards are stochastic, and with probability at least $1 - 13\delta$ when the rewards are adversarial.

⁴ Attaining the no- α -regret property, that is, being no-regret w.r.t. a fraction of the optimum, achieving a competitive ratio, and guaranteeing a fraction of the optimal rewards are used as synonyms in the literature, since any of the aforementioned guarantees can be derived by the others.

We conclude by showing the performance of our algorithm when ρ can be arbitrarily small.

Theorem 7. Suppose that [Condition 2](#) does not hold and the constraints are generated adversarially and are parameterized given B . Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{O}\left(T^{1/4}(\sqrt{T} + B)\right), \quad V_T \leq \tilde{O}(T^{3/4}),$$

with probability at least $1 - 11\delta$ when the rewards are stochastic, and with probability at least $1 - 10\delta$ when the rewards are adversarial.

4.2.2. A weaker baseline

In this section, we show that the impossibility result by Mannor et al. [9] can be circumvented by adopting a different baseline in the regret definition. Precisely, we compute the weaker baseline as the solution to the following linear program:

$$\text{OPT}^w := \begin{cases} \max_{q \in \Delta(M)} & \bar{r}^\top q \\ \text{s.t.} & G_t^\top q \leq 0 \quad \forall t \in [T]. \end{cases}$$

Notice that, in the previous sections, we allow the optimal policy q^* to satisfy the constraints on average, i.e., $\sum_{t=1}^T G_t^\top q^* \leq 0$. In such a case, the set of feasible policies is much smaller than the one associated with the weaker baseline, that is, when a feasible policy must satisfy the constraints *at each episode*. Given the new baseline, we can rewrite the regret as $R_T := T \text{OPT}^w - \sum_{t=1}^T r_t^\top q_t$.

When the regret is computed w.r.t. the weaker baseline, we can recover the same theoretical results of the stochastic setting. Precisely, when [Condition 2](#) holds we have the following result.

Theorem 8.

Suppose that [Condition 2](#) holds and the constraints are generated adversarially. Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{O}(\Lambda \sqrt{T}), \quad V_T \leq \tilde{O}(\Lambda \sqrt{T}),$$

with probability at least $1 - 13\delta$ when the rewards are stochastic, and with probability at least $1 - 12\delta$ when the rewards are adversarial.

We conclude the section by analyzing the scenario in which [Condition 2](#) does not hold.

Theorem 9.

Suppose that [Condition 2](#) does not hold and the constraints are generated adversarially. Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{O}(T^{3/4}), \quad V_T \leq \tilde{O}(T^{3/4}),$$

with probability at least $1 - 12\delta$ when the rewards are stochastic, and with probability at least $1 - 11\delta$ when the rewards are adversarial.

Intuitively, [Theorems 8](#) and [9](#) can be proved by the fact that playing the optimal policy guarantees small violations independently on the episode the optimum is chosen. This is *not* the case for the stronger baseline, since playing the optimum in some episodes may lead to arbitrarily large constraint violations.

5. Conclusions and future works

In this work, we studied online learning in episodic CMDPs under *bandit feedback*, where, after each episode, the learner only observes the realized rewards and costs along the executed trajectory. We introduced the first *best-of-both-worlds* algorithm in this setting, based on a primal-dual policy optimization approach that, differently from Stradi et al. [1], does *not* require repeatedly solving convex programs.

On the theoretical side, when the costs are *stochastic*, our algorithm achieves $\tilde{O}(\sqrt{T})$ regret and constraint violation. When the costs are *adversarial*, it guarantees $\tilde{O}(\sqrt{T})$ violation and a fraction of the optimal reward. These guarantees match those of the full-feedback algorithm by Stradi et al. [1] and are provably tight. We further characterized the performance in terms of a Slater-type parameter ρ : even when ρ is arbitrarily small, we still obtain $\tilde{O}(T^{3/4})$ regret and violation in the stochastic setting, *without* requiring any knowledge of ρ .

Finally, we established refined guarantees in the adversarial regime with *limited adversariality*, obtaining $\tilde{O}(\sqrt{T} + B)$ regret and $\tilde{O}(\sqrt{T})$ violation, where B quantifies non-stationarity, and corresponding bounds when ρ is arbitrarily small. We also showed that sublinear regret and violation are attainable in the adversarial setting when benchmarking against a *weaker* baseline that satisfies the constraints at every round, recovering $\tilde{O}(\sqrt{T})$ rates for sufficiently large ρ and $\tilde{O}(T^{3/4})$ otherwise.

Several directions are left for future investigation. First, it would be interesting to sharpen the dependence on ρ , and to understand whether the $\tilde{O}(T^{3/4})$ regime for arbitrarily small ρ is inherent under bandit feedback, or can be improved under additional structural assumptions. Second, extending our results beyond the tabular setting (e.g., to function approximation and continuous state-action spaces) remains an important open problem. Finally, it would be valuable to study more general feedback models (e.g., delayed feedback) and richer forms of non-stationarity.

CRedit authorship contribution statement

Francesco Emanuele Stradi: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization; **Anna Lunghi:** Investigation, Formal analysis; **Matteo Castiglioni:** Supervision, Investigation, Conceptualization; **Alberto Marchesi:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization; **Nicola Gatti:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

Appendix

The Appendix is structured as follows:

- In [Appendix A](#), we provide additional notation employed in the rest of the appendix.
- In [Appendix B](#), we provide the events dictionary.
- In [Appendix C](#) we provide the theoretical guarantees attained by [Algorithm 3](#). Precisely, we provide the complete version of the primal algorithm (see [Algorithm 4](#)), and analyze the related performances.
- In [Appendix D](#), we provide the theoretical guarantees attained by the dual algorithm.
- In [Appendix E](#), we provide the analysis to bound the Lagrange multipliers during the learning dynamic.
- In [Appendix F](#), we provide the theoretical guarantees attained by [Algorithm 2](#) when the constraints are stochastic.
- In [Appendix G](#), we provide the theoretical guarantees attained by [Algorithm 2](#) when the constraints are adversarial.
- In [Appendix H](#), we provide the theoretical guarantees attained by [Algorithm 2](#) when the constraints are adversarial with limited adversariality.
- In [Appendix I](#) we provide the theoretical guarantees attained by [Algorithm 2](#) when the constraints are adversarial and the baseline is computed w.r.t. the policies that satisfy the constraints at each episode.
- In [Appendix J](#) we provide technical lemmas employed in our work.
- In [Appendix K](#), we provide auxiliary lemmas from existing works.
- In [Appendix L](#), we provide the experimental evaluation of our work.

Appendix A. Additional notation

In the following section, we introduce some useful notation from policy optimization. First, we define the value function $V^\pi(x; f)$, for policy π , state x and generic function f that assumes values for each state $x \in X$ and for each action $a \in A$. Formally,

$$V^\pi(x; f) := \mathbb{E} \left[\sum_{j=h(x)+1}^H f(x_j, a_j) | a_j \sim \pi(\cdot | x_j), x_j \sim P(\cdot | x_{j-1}, a_{j-1}) \right],$$

where $h(x)$ is the layer h such that $x \in X_h$. Notice that the value function can be written using the occupancy measure $q^{\pi, P}$ generated by the policy π and the transition probability P as : $V^\pi(x_0; f) = \sum_{x,a} q^{\pi, P}(x, a) f(x, a)$. We introduce also a Q -function of a generic function f as:

$$\begin{cases} Q(x, a; f) = f(x, a) + \mathbb{E}_{x' \sim P(\cdot | x, a)} [V^\pi(x'; f)] \\ V^\pi(x; f) = \mathbb{E}_{a \sim \pi(\cdot | x)} [Q^\pi(x, a; f)] \\ V^\pi(x_H; f) = 0 \end{cases}$$

In addition we will use the notation $Q_t(x, a)$ to indicate the Q -function computed with respect to the function ℓ_t , i.e. $Q(x, a; \ell_t)$.

Appendix B. Dictionary

In the following, we provide the definition of different quantities which will be employed in the rest of the appendix. This is done for the ease of presentation.

- Quantity $\mathcal{E}_{t_1, t_2}^P$:

$$\mathcal{E}_{t_1, t_2}^P = U_1 \Xi_{t_1, t_2} C \sqrt{T} + U_2 \Xi_{t_1, t_2} \frac{(t_2 - t_1 + 1)}{C \sqrt{T}} + U_3 \Xi_{t_1, t_2} \frac{1}{C \sqrt{T}} + U_4 \Xi_{t_1, t_2} \sqrt{T},$$

where:

$$\begin{aligned}
- U_1 &= 6H^2 \ln \left(\frac{H|A|T^2}{\delta} \right) \\
- U_2 &= 9H|X||A| \\
- U_3 &= \frac{H}{2} \ln \left(\frac{HT^2}{\delta} \right) \\
- U_4 &= 30H^2|X|^2 \sqrt{2|A| \ln \left(\frac{T|X|^2|A|}{\delta} \right)}.
\end{aligned}$$

With probability at least $1 - 4\delta$ it holds $R_{t_1, t_2}^P \leq \mathcal{E}_{t_1, t_2}^P$, $\forall t_1, t_2 \in [T] : 1 \leq t_1 \leq t_2 \leq T$ by [Theorem 1](#).

• **Quantity $\mathcal{E}^D(0)$:**

$$\mathcal{E}^D(0) = D_1 \frac{\|\lambda_{t_1}\|_2^2}{\eta} + D_2 \eta (t_2 - t_1 + 1),$$

where:

$$\begin{aligned}
- D_1 &= \frac{1}{2} \\
- D_2 &= \frac{mH^2}{2}.
\end{aligned}$$

It holds $R_{t_1, t_2}^D(0) \leq \mathcal{E}^D(0)$, $\forall t_1, t_2 \in [T] : 1 \leq t_1 \leq t_2 \leq T$ by [Theorem 11](#).

• **Quantity $\mathcal{E}_{t_1, t_2}^G$:**

$$\mathcal{E}_{t_1, t_2}^G = B_1 \sqrt{(t_2 - t_1 + 1)},$$

where:

$$- B_1 = 2H \sqrt{\ln \left(\frac{T^2}{\delta} \right)}.$$

Given a $q \in \Delta(M)$, with probability at least $1 - \delta$ it holds in case of stochastic constraints $\sum_{t=t_1}^{t_2} (G_t^\top q - \bar{G}^\top q) \leq \mathcal{E}_{t_1, t_2}^G$, by Azuma-Hoeffding inequality.

• **Quantity $\mathcal{E}_{t_1, t_2}^I$:**

$$\mathcal{E}_{t_1, t_2}^I = F_1 \sqrt{(t_2 - t_1 + 1)},$$

where:

$$- F_1 = H \sqrt{2 \ln \left(\frac{T^2}{\delta} \right)}.$$

With probability at least $1 - \delta$ it holds $\sum_{t=t_1}^{t_2} \sum_{x,a} (\mathbb{I}_t(x, a) - q_t(x, a)) \leq \mathcal{E}_{t_1, t_2}^I$, and with probability at least $1 - \delta$ it holds $\sum_{t=t_1}^{t_2} \sum_{x,a} (q_t(x, a) - \mathbb{I}_t(x, a)) \leq \mathcal{E}_{t_1, t_2}^I$, by Azuma-Hoeffding inequality.

• **Quantity C :**

$$C = 252|X||A|H$$

• **Quantity D :**

$$\begin{aligned}
D &= 84672mH^2|X|^2|A| \\
&= 336mH|X|C.
\end{aligned}$$

Appendix C. Omitted proofs for the primal algorithm

In this section we study the guarantees attained by the primal procedure, namely [Algorithm 4](#).

Theorem 10. For any $\delta \in (0, 1)$, [Algorithm 4](#) attains, with probability at least $1 - 4\delta$ and for all $[t_1, \dots, t_2] \subset [T]$:

$$\begin{aligned}
& \sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a (\pi_t(a|x) - \pi^*(a|x)) (Q_t^{\pi_t}(x, a) - B_t(x, a)) \\
&= \Xi_{t_1, t_2} o(T) + \sum_{t=t_1}^{t_2} V^{\pi^*}(x_0; b_t) + \frac{1}{H} \sum_{t=t_1}^{t_2} \sum_{x,a} q^*(x) \pi_t(a|x) B_t(x, a),
\end{aligned}$$

for all $t_1, t_2 \in [T]$ s.t. $1 \leq t_1 \leq t_2 \leq T$ and where $\Xi_{t_1, t_2} \geq \max_{t \in [t_1, \dots, t_2]} \max_{x,a} \ell_t(x, a)$.

Proof. In the rest of the proof, we will refer as $\bar{L}_t$ to $\max_{\tau \in [t]} \max_{h \in [H]} L_{\tau, h}$ and $\bar{L}_{t_1, t_2}$ to $\max_{\tau \in [t_1, \dots, t_2]} \max_{h \in [H]} L_{\tau, h}$; therefore, by definition it holds $\bar{L}_t \leq H\Xi_t$ for all $t \in [T]$.

As a first step, we decompose $\sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a (\pi_t(a|x) - \pi^*(a|x)) (Q_t^{\pi_t}(x, a) - B_t(x, a))$ in three different quantities:

$$\sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a (\pi_t(a|x) - \pi^*(a|x)) (Q_t^{\pi_t}(x, a) - B_t(x, a))$$

Algorithm 4 FS-PODB.**Require:** $X, A, \sigma = \frac{1}{T}, C$

- 1: $\mathcal{P}_1 \leftarrow$ set of all possible transitions
- 2: $\pi_1(a|x) = \frac{1}{|A|} \quad \forall (x, a) \in X \times A$
- 3: $\Xi_0 \leftarrow 1$
- 4: $\gamma \leftarrow \frac{1}{C\sqrt{T}}$
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Play π_t , observe $\{(x_h, a_h)\}_{h=0}^{H-1}$, losses $\{\ell_t(x_h, a_h)\}_{h=0}^{H-1}$ and Ξ_t
- 7: $\eta_t \leftarrow \frac{1}{2H\Xi_t C\sqrt{T}}$
- 8: For all $h = 0, \dots, H-1$ and $(x, a) \in X_h \times A$:

$$L_{t,h} = \sum_{j=h}^{H-1} \ell_t(x_h, a_h)$$

$$\hat{Q}_t(x, a) = \frac{L_{t,h}}{\bar{q}_t(x, a) + \gamma} \mathbb{I}_t(x, a),$$

where $\bar{q}_t(x, a) = \max_{\hat{P} \in \mathcal{P}_t} q^{\hat{P}, \pi_t}(x, a)$ and $\mathbb{I}_t(x, a) = \mathbb{I}\{x_{t,h} = x, a_{t,h} = a\}$.

- 9: For all $(x, a) \in X \times A$:

$$b_t(x) = \mathbb{E}_{a \sim \pi_t(\cdot|x)} \left[\frac{3\gamma H \Xi_t + H \Xi_t (\bar{q}_t(x, a) - \underline{q}_t(x, a))}{\bar{q}_t(x, a) + \gamma} \right]$$

$$B_t(x, a) = b_t(x) + \left(1 + \frac{1}{H}\right) \max_{\hat{P} \in \mathcal{P}_t} \mathbb{E}_{x' \sim \hat{P}(\cdot|x, a)} \mathbb{E}_{a' \sim \pi_t(\cdot|x')} [B_t(x', a')]$$

where $\underline{q}_t = \min_{\hat{P} \in \mathcal{P}_t} q^{\hat{P}, \pi_t}(x, a)$, and $B_t(x_H, a) = 0$ for all a .

- 10: For all $(x, a) \in X \times A$:

$$w_{t+1}(a|x) = (1 - \sigma) w_t(a|x) e^{-\eta_t (\hat{Q}_t(x, a) - B_t(x, a))} + \frac{\sigma}{|A|} \sum_{a' \in A} w_t(a'|x) e^{-\eta_t (\hat{Q}_t(x, a') - B_t(x, a'))}.$$

$$\pi_{t+1}(a|x) = \frac{w_{t+1}(a|x)}{\sum_{a' \in A} w_{t+1}(a'|x)}.$$

- 11: $\mathcal{P}_{t+1} \leftarrow \text{TRANSITION.UPDATE}(\{(x_h, a_h)\}_{h=0}^{H-1})$

$$\begin{aligned}
&= \underbrace{\sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a (\pi_t(a|x) - \pi^*(a|x)) (\hat{Q}_t(x, a) - B_t(x, a))}_{\textcircled{1}} \\
&\quad + \underbrace{\sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a \pi_t(a|x) (Q_t^{\pi_t}(x, a) - \hat{Q}_t(x, a))}_{\textcircled{2}} \\
&\quad + \underbrace{\sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a \pi^*(a|x) (\hat{Q}_t(x, a) - Q_t^{\pi_t}(x, a))}_{\textcircled{3}},
\end{aligned}$$

which we proceed to bound separately.

Bound on ①. The quantity of interest can be bounded after noticing that [Algorithm 4](#) employs a slightly modified version of OMD. In fact, recalling the definition of π_t , we can write:

$$\pi_{t+1}(a|x) = \frac{w_{t+1}(a|x)}{\sum_{a'} w_{t+1}(a'|x)}$$

$$\begin{aligned}
&= \frac{(1-\sigma)w_t(a|x)e^{-\eta_t(\hat{Q}_t(x,a)-B_t(x,a))} + \frac{\sigma}{|A|} \sum_{a' \in A} w_t(a'|x)e^{-\eta_t(\hat{Q}_t(x,a')-B_t(x,a'))}}{\sum_{a' \in A} w_t(a'|x)e^{-\eta_t(\hat{Q}_t(x,a')-B_t(x,a'))}} \\
&= (1-\sigma) \frac{\pi_t(a|x)e^{-\eta_t(\hat{Q}_t(x,a)-B_t(x,a))}}{\sum_{a' \in A} \pi_t(a'|x)e^{-\eta_t(\hat{Q}_t(x,a')-B_t(x,a'))}} + \sigma \frac{1}{|A|}.
\end{aligned}$$

From now on we will refer to $\frac{\pi_t(a|x)e^{-\eta_t(\hat{Q}_t(x,a)-B_t(x,a))}}{\sum_{a' \in A} \pi_t(a'|x)e^{-\eta_t(\hat{Q}_t(x,a')-B_t(x,a'))}}$ as $\tilde{\pi}_{t+1}(x, a)$. Thus,

$$\pi_{t+1}(a|x) = (1-\sigma)\tilde{\pi}_{t+1}(x, a) + \frac{\sigma}{|A|}.$$

Calling $\psi(\cdot)$ the negative entropy function defined as $\psi(\pi(\cdot|x)) := \sum_a \pi(a|x) \ln(\pi(a|x))$, by standard analysis (e.g. [38]), it holds:

$$\tilde{\pi}_{t+1}(\cdot|x) = \arg \min_{\pi(\cdot|x) \in \Delta(A)} \sum_a \left(\hat{Q}_t(x, a) - B_t(x, a) \right) \pi(a|x) + \frac{1}{\eta} D_\psi(\pi(\cdot|x); \pi_t(\cdot|x)),$$

where D_ψ is Bregman divergence w.r.t. the negative entropy function $\psi(\cdot)$. Thus, for all $\pi(\cdot|x)$ it holds $\eta_t \sum_a \left(\hat{Q}_t(x, a) - B_t(x, a) \right) (\pi(a|x) - \tilde{\pi}_{t+1}(x, a)) + \langle \nabla \psi(\tilde{\pi}_{t+1}(\cdot|x)) - \nabla \psi(\pi_t(\cdot|x)), \pi(\cdot|x) - \tilde{\pi}_{t+1}(\cdot|x) \rangle \geq 0$.

So, for all $\pi(\cdot|x)$ the following holds:

$$\begin{aligned}
&\eta_t \langle \hat{Q}_t(x, \cdot) - B_t(x, \cdot), \pi_t(\cdot|x) - \pi(\cdot|x) \rangle \\
&= \eta_t \langle \hat{Q}_t(x, \cdot) - B_t(x, \cdot) + \nabla \psi(\tilde{\pi}_{t+1}(\cdot|x)) - \nabla \psi(\pi_t(\cdot|x)), \tilde{\pi}_{t+1}(\cdot|x) - \pi(\cdot|x) \rangle \\
&\quad + \eta_t \langle \hat{Q}_t(x, \cdot) - B_t(x, \cdot), \pi_t(\cdot|x) - \tilde{\pi}_{t+1}(\cdot|x) \rangle \\
&\quad + \langle \nabla \psi(\tilde{\pi}_{t+1}(\cdot|x)) - \nabla \psi(\pi_t(\cdot|x)), \pi(\cdot|x) - \tilde{\pi}_{t+1}(\cdot|x) \rangle \\
&\leq \langle \eta_t (\hat{Q}_t(x, \cdot) - B_t(x, \cdot)), \pi_t(\cdot|x) - \tilde{\pi}_{t+1}(\cdot|x) \rangle \\
&\quad + \langle \nabla \psi(\tilde{\pi}_{t+1}(\cdot|x)) - \nabla \psi(\pi_t(\cdot|x)), \pi(\cdot|x) - \tilde{\pi}_{t+1}(\cdot|x) \rangle \\
&\leq D_\psi(\pi_t(\cdot|x); \pi_t(\cdot|x)) - D_\psi(\pi_t(\cdot|x); \tilde{\pi}_{t+1}(\cdot|x)) - D_\psi(\tilde{\pi}_{t+1}(\cdot|x); \pi_t(\cdot|x)) \\
&\quad + \eta_t \langle \hat{Q}_t(x, \cdot) - B_t(x, \cdot), \pi_t(\cdot|x) - \tilde{\pi}_{t+1}(\cdot|x) \rangle \tag{C.1}
\end{aligned}$$

$$= D_\psi(\pi_t(\cdot|x); \pi_t(\cdot|x)) - D_\psi(\pi_t(\cdot|x); \tilde{\pi}_{t+1}(\cdot|x)) + \frac{\eta_t^2}{2} \sum_{a \in A} \left(\hat{Q}_t(x, a) - B_t(x, a) \right)^2 \pi_t(a|x), \tag{C.2}$$

where Inequality (C.1) and Inequality (C.2) are based on the proofs of Lemma 6.6. and Lemma 6.9. in [38].

Additionally we can show that for all $t \in [T]$: $D_\psi(\pi(\cdot|x); \pi_t(\cdot|x)) - D_\psi(\pi(\cdot|x); \tilde{\pi}_t(\cdot|x)) \leq \sigma \ln(|A|)$. Indeed,

$$\begin{aligned}
&D_\psi(\pi(\cdot|x); \pi_t(\cdot|x)) - D_\psi(\pi(\cdot|x); \tilde{\pi}_t(\cdot|x)) \\
&= D_\psi \left(\pi(\cdot|x); (1-\sigma)\tilde{\pi}_t(\cdot|x) + \sigma \pi \frac{1}{|A|} \right) - D_\psi(\pi(\cdot|x); \tilde{\pi}_t(\cdot|x)) \\
&\leq \sigma D_\psi \left(\pi(\cdot|x); \pi \frac{1}{|A|} \right) - \sigma D_\psi(\pi(\cdot|x); \tilde{\pi}_t(\cdot|x)) \\
&\leq \sigma \ln(|A|),
\end{aligned}$$

where the last inequality holds since $D_\psi(\pi(\cdot|x); \tilde{\pi}_t(\cdot|x)) \geq 0$ and

$$\begin{aligned}
D_\psi(\pi(\cdot|x); \pi \frac{1}{|A|}) &= \sum_{a \in A} \pi(a|x) \ln \left(\frac{\pi(a|x)}{\pi \frac{1}{|A|}(a|x)} \right) \\
&\leq \sum_{a \in A} \pi(a|x) \ln \left(\frac{1}{\pi \frac{1}{|A|}(a|x)} \right) \\
&= \sum_{a \in A} \pi(a|x) \ln(|A|) \\
&= \ln(|A|).
\end{aligned}$$

Notice that with we refer as $\pi \frac{1}{|A|}$ to the vector strategy in $[0, 1]^{|A|}$ with all elements equal to $\frac{1}{|A|}$.

Moreover we bound $D_\psi(\pi(\cdot|x); \pi_{t_1}(\cdot|x))$, since $\pi_{t_1}(a|x) = (1-\sigma)\tilde{\pi}_{t_1}(a|x) + \sigma(\frac{1}{|A|}) \geq \frac{\sigma}{|A|}$, as follows:

$$D_\psi(\pi(\cdot|x); \pi_{t_1}(\cdot|x)) = \sum_{a \in A} \pi(a|x) \ln \left(\frac{\pi(a|x)}{\pi_{t_1}(a|x)} \right)$$

$$\begin{aligned}
&\leq \sum_{a \in A} \pi(a|x) \ln \left(\frac{1}{\pi_{t_1}(a|x)} \right) \\
&\leq \sum_{a \in A} \pi(a|x) \ln \left(\frac{|A|}{\sigma} \right) \\
&= \ln \left(\frac{|A|}{\sigma} \right).
\end{aligned}$$

Putting everything together we have that:

$$\begin{aligned}
\textcircled{1} &= \sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a (\pi_t(a|x) - \pi^*(a|x)) \left(\hat{Q}_t(x, a) - B_t(x, a) \right) \\
&\leq \sum_x q^*(x) \left(\frac{D_\psi(\pi(\cdot|x); \pi_{t_1}(\cdot|x))}{\eta_{t_1}} + \sum_{t=t_1+1}^{t_2} \left(\frac{D_\psi(\pi(\cdot|x); \pi_t(\cdot|x))}{\eta_t} - \frac{D_\psi(\pi(\cdot|x); \tilde{\pi}_t(\cdot|x))}{\eta_{t+1}} \right) \right) \\
&\quad + \sum_x q^*(x) \sum_{t=t_1}^{t_2} \frac{\eta_t}{2} \left(\hat{Q}_t(x, a) - B_t(x, a) \right)^2 \pi_t(a|x) \tag{C.3a}
\end{aligned}$$

$$\begin{aligned}
&\leq \sum_x q^*(x) \left(\frac{D_\psi(\pi(\cdot|x); \pi_{t_1}(\cdot|x))}{\eta_{t_1}} + \sum_{t=t_1+1}^{t_2} \left(\frac{D_\psi(\pi(\cdot|x); \pi_t(\cdot|x)) - D_\psi(\pi(\cdot|x); \tilde{\pi}_t(\cdot|x))}{\eta_t} \right) \right) \\
&\quad + \sum_x q^*(x) \sum_{t=t_1}^{t_2} \frac{\eta_t}{2} \left(\hat{Q}_t(x, a) - B_t(x, a) \right)^2 \pi_t(a|x) \tag{C.3b}
\end{aligned}$$

$$\leq \frac{\ln \left(\frac{|A|}{\sigma} \right)}{\eta_{t_1}} + \sigma \sum_{t=t_1+1}^{t_2} \frac{\ln(|A|)}{\eta_{t_2}} + \sum_x q^*(x) \sum_{t=t_1}^{t_2} \frac{\eta_t}{2} \left(\hat{Q}_t(x, a) - B_t(x, a) \right)^2 \pi_t(a|x) \tag{C.3c}$$

$$\begin{aligned}
&\leq \frac{\ln \left(\frac{|A|}{\sigma} \right)}{\eta_{t_1}} + \frac{\sigma T \ln(|A|)}{\eta_{t_2}} + \sum_x q^*(x) \sum_{t=t_1}^{t_2} \frac{\eta_t}{2} \left(\hat{Q}_t(x, a) - B_t(x, a) \right)^2 \pi_t(a|x) \\
&= \frac{\ln(|A|T)}{\eta_{t_1}} + \frac{\ln(|A|)}{\eta_{t_2}} + \sum_x q^*(x) \sum_{t=t_1}^{t_2} \frac{\eta_t}{2} \left(\hat{Q}_t(x, a) - B_t(x, a) \right)^2 \pi_t(a|x),
\end{aligned}$$

where $\sigma = \frac{1}{T}$, Inequality (C.3a) holds by Inequality (C.2), Inequality (C.3b) holds since $\frac{1}{\eta_{t+1}} \geq \frac{1}{\eta_t}$ for all $t \in [T]$, and Inequality (C.3c) holds since $\eta_{t_2} \leq \eta_t$ for all t in $[t_1 + 1, \dots, t_2]$. Focusing now on the last part of the right term, with probability at least $1 - 2\delta$ the following holds:

$$\begin{aligned}
&\sum_x \sum_a q^*(x) \sum_{t=t_1}^{t_2} \frac{\eta_t}{2} \left(\hat{Q}_t(x, a) - B_t(x, a) \right)^2 \pi_t(a|x) \\
&\leq \sum_{t=t_1}^{t_2} \eta_t \sum_x \sum_a q^*(x) \pi_t(a|x) \hat{Q}_t(x, a)^2 + \sum_{t=t_1}^{t_2} \eta_t \sum_x \sum_a q^*(x) \pi_t(a|x) B_t(x, a)^2 \tag{C.4a}
\end{aligned}$$

$$= \sum_{t=t_1}^{t_2} \eta_t \sum_x \sum_a q^*(x) \pi_t(a|x) \frac{L_{t,h}^2}{(\bar{q}_t(x, a) + \gamma)^2} \mathbb{I}_t(x, a) + \sum_{t=t_1}^{t_2} \eta_t \sum_x \sum_a q^*(x) \pi_t(a|x) B_t(x, a)^2 \tag{C.4b}$$

$$\leq \bar{L}_{t_1, t_2} \sum_{t=t_1}^{t_2} \eta_t \bar{L}_t \sum_x \sum_a \frac{q^*(x) \pi_t(a|x)}{\bar{q}_t(x, a) + \gamma} \frac{\mathbb{I}_t(x, a)}{\bar{q}_t(x, a) + \gamma} + \sum_{t=t_1}^{t_2} \eta_t \sum_x \sum_a q^*(x) \pi_t(a|x) B_t(x, a)^2 \tag{C.4c}$$

$$\begin{aligned}
&\leq \frac{\gamma}{2H} \bar{L}_{t_1, t_2} \sum_{t=t_1}^{t_2} \sum_x \sum_a \frac{q^*(x) \pi_t(a|x)}{\bar{q}_t(x, a) + \gamma} \frac{q_t(x, a)}{\bar{q}_t(x, a) + \gamma} + \frac{\gamma \bar{L}_{t_1, t_2}}{2} \ln \left(\frac{HT^2}{\delta} \right) \\
&\quad + \sum_{t=t_1}^{t_2} \eta_t \sum_x \sum_a q^*(x) \pi_t(a|x) B_t(x, a)^2 \tag{C.4d}
\end{aligned}$$

$$\begin{aligned}
&\leq \sum_{t=t_1}^{t_2} \sum_x \sum_a q^*(x) \pi_t(a|x) \frac{\gamma \Xi_{t_1, t_2}}{2(\bar{q}_t(x, a) + \gamma)} + \frac{\gamma \bar{L}_{t_1, t_2}}{2} \ln \left(\frac{HT^2}{\delta} \right) \\
&\quad + \frac{1}{2H} \sum_{t=t_1}^{t_2} \sum_x \sum_a q^*(x) \pi_t(a|x) B_t(x, a) \tag{C.4e}
\end{aligned}$$

$$= \sum_{t=t_1}^{t_2} \sum_x \sum_a q^*(x) \pi_t(a|x) \left(\frac{\gamma \Xi_{t_1, t_2}}{2(\bar{q}_t(x, a) + \gamma)} + \frac{B_t(x, a)}{2H} \right) + \frac{\gamma \bar{L}_{t_1, t_2}}{2} \ln \left(\frac{HT^2}{\delta} \right),$$

where Inequality (C.4a) holds since $(a - b)^2 \leq 2a^2 + 2b^2$, for all $a, b \in \mathbb{R}$, Equality (C.4b) holds by definition of $\hat{Q}(x, a)$, Inequality (C.4c) is motivated by the fact that $L_{t,h} \leq \bar{L}_{t_1,t_2}$ by its definition, Inequality (C.4d) holds with probability at least $1 - \delta$ by applying Lemma 9 and taking $\alpha_t(x, a) = \frac{q^*(x)\pi_t(a|x)}{\bar{q}_t(x,a)+\gamma}$ since $\frac{q^*(x)\pi_t(a|x)}{\bar{q}_t(x,a)+\gamma} \leq \frac{1}{\gamma}$ and considering that by definition $\eta_t \Xi_t = \frac{\gamma}{2H}$, and finally Inequality (C.4e) holds since $\bar{q}_t(x, a) \geq q_t(x, a)$, $\forall (x, a) \in X \times A, \forall t \in [t_1 \dots t_2]$ with probability at least $1 - \delta$ by Lemma 12 and by Lemma 6. Setting $\gamma = 2\eta_t H \Xi_t$, we can conclude that, with probability at least $1 - 2\delta$, ① is bounded as:

$$\frac{4H\Xi_{t_1,t_2} \ln(|A|T)}{\gamma} + \gamma \frac{H\Xi_{t_1,t_2}}{2} \ln \left(\frac{HT^2}{\delta} \right) + \sum_{t=t_1}^{t_2} \sum_x \sum_a q^*(x)\pi_t(a|x) \left(\frac{\gamma\Xi_{t_1,t_2}}{2(\bar{q}_t(x, a) + \gamma)} + \frac{B_t(x, a)}{2H} \right).$$

Bound on ②. To bound ② we employ the same approach as in [19]. First we define Y_t as $Y_t := \sum_x \sum_a q^*(x)\pi_t(a|x)\hat{Q}_t(x, a)$, for all $t \in [T]$. Now since $\sum_{t=t_1}^{t_2} Y_t$ is a martingale sequence, we apply Freedman's inequality. First notice that under the event $P \in \mathcal{P}_t(t)$ for all $t \in [T]$:

$$\begin{aligned} \mathbb{E}[Y_t^2] &\leq \mathbb{E}_t \left[\left(\sum_x \sum_a q^*(x)\pi_t(a|x)\hat{Q}_t(x, a) \right)^2 \right] \\ &\leq \mathbb{E}_t \left[\left(\sum_x \sum_a q^*(x)\pi_t(a|x) \right) \left(\sum_x \sum_a q^*(x)\pi_t(a|x)\hat{Q}_t(x, a)^2 \right) \right] \\ &= H \mathbb{E}_t \left[\sum_x \sum_a q^*(x)\pi_t(a|x)\hat{Q}_t(x, a)^2 \right] \\ &= H \sum_x \sum_a q^*(x)\pi_t(a|x) \frac{L_{t,h}^2}{(\bar{q}_t(x, a) + \gamma)^2} q_t(x, a) \\ &\leq \sum_x \sum_a q^*(x)\pi_t(a|x) \frac{H\bar{L}_t^2}{\bar{q}_t(x, a) + \gamma}. \end{aligned}$$

Thus, thanks to Lemma 8, since $|Y_t| \leq H \sup_{x,a'} \hat{Q}_t(x, a) \leq \frac{H\bar{L}_t}{\gamma}$, with probability at least $1 - \delta$ it holds simultaneously for all $t_1, t_2 : 1 \leq t_1 \leq t_2 \leq T$:

$$\sum_{t=t_1}^{t_2} (\mathbb{E}_t[Y_t] - Y_t) \leq \frac{\gamma}{H\bar{L}_{t_1,t_2}} \sum_{t=t_1}^{t_2} \sum_x \sum_a q^*(x)\pi_t(a|x) \frac{H\bar{L}_t^2}{\bar{q}_t(x, a) + \gamma} + \frac{H\bar{L}_{t_1,t_2}}{\gamma} \log \left(\frac{T^2}{\delta} \right).$$

We notice also the following result with probability at least $1 - \delta$ for all $t \in [T]$:

$$\begin{aligned} &\sum_x \sum_a q^*(x)\pi_t(a|x)Q_t(x, a) - \mathbb{E}[Y_t] \\ &= \sum_x \sum_a q^*(x)\pi_t(a|x)Q_t(x, a) - \mathbb{E} \left[\sum_x \sum_a q^*(x)\pi_t(a|x)\hat{Q}_t(x, a) \right] \\ &= \sum_x \sum_a q^*(x)\pi_t(a|x)Q_t(x, a) \left(1 - \frac{q_t(x, a)}{\bar{q}_t(x, a) + \gamma} \right) \\ &\leq \sum_x \sum_a q^*(x)\pi_t(a|x)H\Xi_t \left(\frac{\bar{q}_t(x, a) - q_t(x, a) + \gamma}{\bar{q}_t(x, a) + \gamma} \right) \\ &\leq \sum_x \sum_a q^*(x)\pi_t(a|x)H\Xi_t \left(\frac{(\bar{q}_t(x, a) - \underline{q}_t(x, a)) + \gamma}{\bar{q}_t(x, a) + \gamma} \right). \end{aligned}$$

Finally we can bound ② with probability at least $1 - 2\delta$ as follows.

$$\begin{aligned} \textcircled{2} &= \sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a \pi_t(a|x) \left(Q_t^{\pi_t}(x, a) - \hat{Q}_t(x, a) \right) \\ &= \sum_{t=t_1}^{t_2} (\mathbb{E}_t[Y_t] - Y_t) + \sum_{t=t_1}^{t_2} \left(\sum_x \sum_a q^*(x)\pi_t(a|x)Q_t(x, a) - \mathbb{E}[Y_t] \right) \\ &\leq \sum_{t=t_1}^{t_2} \sum_x \sum_a q^*(x)\pi_t(a|x)H\Xi_t \left(\frac{(\bar{q}_t(x, a) - \underline{q}_t(x, a)) + 2\gamma}{\bar{q}_t(x, a) + \gamma} \right) + \frac{H\bar{L}_{t_1,t_2}}{\gamma} \ln \left(\frac{T^2}{\delta} \right). \end{aligned}$$

Bound on ③. With probability at least $1 - 2\delta$ it holds:

$$\textcircled{3} = \sum_x q^*(x) \sum_{t=t_1}^{t_2} \sum_a \pi^*(a|x) \left(\hat{Q}_t(x, a) - Q_t^{\pi_t}(x, a) \right) \leq \frac{H^2\Xi_{t_1,t_2}}{2\gamma} \ln \left(\frac{HT^2}{\delta} \right),$$

by Corollary 1.

Conclusion of the proof. Finally we notice that, with probability at least $1 - 4\delta$, we have the following result.

$$\begin{aligned} \sum_x q^*(x) \sum_{t=1}^{t_2} \sum_a (\pi_t(a|x) - \pi^*(a|x)) (Q_t^{\pi_t}(x, a) - B_t(x, a)) &= \textcircled{1} + \textcircled{2} + \textcircled{3} \\ &\leq \gamma \frac{H\Xi_{t_1, t_2}}{2} \ln \left(\frac{HT^2}{\delta} \right) + \frac{6H^2\Xi_{t_1, t_2}}{\gamma} \ln \left(\frac{H|A|T^2}{\delta} \right) \\ &\quad + \sum_{t=1}^{t_2} \sum_{x,a} q^*(x) \pi_t(a|x) \left(\frac{\Xi_t(3\gamma H + H(\bar{q}_t(x, a) - q_t(x, a)))}{\bar{q}_t(x, a) + \gamma} + \frac{B_t(x, a)}{H} \right). \end{aligned}$$

This concludes the proof. $\square$

Theorem 1. For any $\delta \in (0, 1)$, with probability at least $1 - 8\delta$, Algorithm FS-PODB attains:

$$R_{t_1, t_2} \leq \tilde{\mathcal{O}} \left(\Xi_{t_1, t_2} \sqrt{T} + \Xi_{t_1, t_2} \frac{t_2 - t_1}{\sqrt{T}} \right),$$

where the regret can be computed with respect to any policy function $\pi : X \rightarrow \Delta(A)$.

Proof. By means of Theorem 10 and by Lemma 7 we have that with probability at least $1 - 4\delta$:

$$R_{t_1, t_2}^P \leq \gamma \frac{H\Xi_{t_1, t_2}}{2} \ln \left(\frac{HT^2}{\delta} \right) + \frac{6H^2\Xi_{t_1, t_2}}{\gamma} \ln \left(\frac{H|A|T^2}{\delta} \right) + 3 \sum_{t=1}^{t_2} \hat{V}^{\pi_t}(x_0; b_t).$$

We can bound $\sum_{t=1}^T \hat{V}^{\pi_t}(x_0; b_t)$, with probability at least $1 - 4\delta$, as:

$$\begin{aligned} &\sum_{t=1}^{t_2} \hat{V}^{\pi_t}(x_0; b_t) \\ &= \sum_{t=1}^{t_2} \sum_{x,a} q^{\hat{\pi}_t, \pi_t}(x, a) \left(\frac{H\Xi_t(\bar{q}_t(x, a) - q_t(x, a)) + 3H\Xi_t\gamma}{\bar{q}_t(x, a) + \gamma} \right) \\ &\leq \sum_{t=1}^{t_2} \sum_{x,a} H\Xi_t (\bar{q}_t(x, a) - q_t(x, a) + 3\gamma) \\ &\leq \sum_{t=1}^{t_2} \sum_{x,a} H\Xi_t (\bar{q}_t(x, a) - q_t(x, a)) + 3\Xi_{t_1, t_2} \gamma H(t_2 - t_1 + 1) |X| |A| \\ &\leq 4H^2\Xi_{t_1, t_2} |X|^2 \sqrt{2T \ln \left(\frac{H|X|}{\delta} \right)} + 6\Xi_{t_1, t_2} H^2 |X|^2 \sqrt{2T |A| \ln \left(\frac{T|X|^2 |A|}{\delta} \right)} \\ &\quad + 3\Xi_{t_1, t_2} \gamma H |X| |A| (t_2 - t_1 + 1), \end{aligned}$$

where the second inequality holds under the event $q^{\hat{\pi}_t, \pi_t}(x, a) \leq \bar{q}_t(x, a)$ for all $(x, a) \in X \times A$ and for all $t \in [T]$ and the last inequality uses Lemma 10.

Thus, with probability at least $1 - 8\delta$, it holds:

$$\begin{aligned} R_{t_1, t_2}^P &\leq \gamma \frac{H\Xi_{t_1, t_2}}{2} \ln \left(\frac{HT^2}{\delta} \right) + \frac{6H^2\Xi_{t_1, t_2}}{\gamma} \ln \left(\frac{H|A|T^2}{\delta} \right) + 30\Xi_{t_1, t_2} H^2 |X|^2 \sqrt{2T |A| \ln \left(\frac{T|X|^2 |A|}{\delta} \right)} + \\ &\quad + 9\Xi_{t_1, t_2} \gamma H |X| |A| (t_2 - t_1 + 1) \\ &\leq \frac{H\Xi_{t_1, t_2}}{2C\sqrt{T}} \ln \left(\frac{HT^2}{\delta} \right) + 6H^2\Xi_{t_1, t_2} C\sqrt{T} \ln \left(\frac{H|A|T^2}{\delta} \right) \\ &\quad + 30\Xi_{t_1, t_2} H^2 |X|^2 \sqrt{2T |A| \ln \left(\frac{T|X|^2 |A|}{\delta} \right)} + 9H |X| |A| \Xi_{t_1, t_2} \frac{(t_2 - t_1 + 1)}{C\sqrt{T}} \\ &= U_1 \Xi_{t_1, t_2} C\sqrt{T} + U_2 \Xi_{t_1, t_2} \frac{(t_2 - t_1 + 1)}{C\sqrt{T}} + U_3 \Xi_{t_1, t_2} \frac{1}{C\sqrt{T}} + U_4 \Xi_{t_1, t_2} \sqrt{T} \\ &= \mathcal{E}_{t_1, t_2}^P, \end{aligned}$$

which concludes the proof. $\square$

Appendix D. Omitted proofs for the dual algorithm

Theorem 11. When employed by [Algorithm 2](#), online projected gradient descent (OGD) attains:

$$R_{t_1, t_2}^D(\lambda) = \sum_{t=t_1}^{t_2} (\lambda - \lambda_t)^\top \sum_{h=0}^{H-1} G_t(x_h, a_h) \leq \frac{\|\lambda_{t_1} - \lambda\|_2^2}{2\eta} + \frac{\eta}{2}(t_2 - t_1 + 1)mH^2.$$

Proof. We proceed to prove the theorem following [\[38\]](#). Indeed, it holds:

$$\begin{aligned} R_{t_1, t_2}^D(\lambda) &= \sum_{t=t_1}^{t_2} (\lambda - \lambda_t)^\top \sum_{h=0}^{H-1} G_t(x_h, a_h) \\ &\leq \frac{\|\lambda_{t_1} - \lambda\|_2^2}{2\eta} + \frac{\eta}{2} \sum_{t=t_1}^{t_2} \left\| \sum_{h=0}^{H-1} G_t(x_h, a_h) \right\|_2^2 \\ &\leq \frac{\|\lambda_{t_1} - \lambda\|_2^2}{2\eta} + \frac{\eta}{2} \sum_{t=t_1}^{t_2} \sum_{i=1}^m \left(\sum_{h=0}^{H-1} G_{t,i}(x_h, a_h) \right)^2 \\ &\leq \frac{\|\lambda_{t_1} - \lambda\|_2^2}{2\eta} + \frac{\eta}{2} \sum_{t=t_1}^{t_2} mH^2 \\ &\leq \frac{\|\lambda_{t_1} - \lambda\|_2^2}{2\eta} + \frac{\eta}{2}(t_2 - t_1 + 1)mH^2. \end{aligned}$$

This concludes the proof. $\square$

Lemma 1. When employed by [Algorithm 2](#), online projected gradient descent (OGD) guarantees for all $t \in [T]$:

$$\|\lambda_{t+1}\|_1 - \|\lambda_t\|_1 \leq mH\eta.$$

Proof. It holds:

$$\begin{aligned} \lambda_{t+1,i} &= \min \left\{ \max \left\{ 0, \lambda_{t,i} + \eta \sum_{h=0}^{H-1} g_{t,i}(x_h, a_h) \right\}, T^{\frac{1}{4}} \right\} \\ &\leq \max \left\{ 0, \lambda_{t,i} + \eta \sum_{h=0}^{H-1} g_{t,i}(x_h, a_h) \right\} \\ &\leq \max \left\{ 0, \lambda_{t,i} + \eta \sum_{h=0}^{H-1} 1 \right\} \\ &= \lambda_{t,i} + \eta H, \end{aligned}$$

which concludes the proof when we take the sum over all $i \in [m]$. $\square$

Appendix E. Analysis on Lagrangian multipliers

Lemma 2. The loss given to the primal algorithm at episode $t \in [T]$, which is defined as $\ell_t(x, a) = \Gamma_t + \sum_{i \in [m]} \lambda_{t,i} g_{t,i}(x, a) - r_t(x, a)$, is such that, considering the Lagrangian loss function $\ell_t^{\mathcal{L}}(x, a) = \sum_{i \in [m]} \lambda_{t,i} g_{t,i}(x, a) - r_t(x, a)$, it holds:

$$\ell_t^\top(q_t - q^*) = \ell_t^{\mathcal{L}, \top}(q_t - q^*),$$

and additionally, ℓ_t assume values in the bounded interval $[0, \Xi_t]$.

Proof. By simple computation, it holds:

$$\begin{aligned} &\ell_t^\top(q_t - q^*) - \ell_t^{\mathcal{L}, \top}(q_t - q^*) \\ &= \left(\sum_{x,a} \Gamma_t(q_t(x, a) - q^*(x, a)) + \sum_{i \in [m]} \sum_{x,a} \lambda_{t,i} g_{t,i}(x, a)(q_t(x, a) - q^*(x, a)) \right. \\ &\quad \left. - \sum_{x,a} r_t(x, a)(q_t(x, a) - q^*(x, a)) \right) \\ &\quad - \left(\sum_{i \in [m]} \sum_{x,a} \lambda_{t,i} g_{t,i}(x, a)(q_t(x, a) - q^*(x, a)) - \sum_{x,a} r_t(x, a)(q_t(x, a) - q^*(x, a)) \right) \\ &= \sum_{x,a} \Gamma_t(q_t(x, a) - q^*(x, a)) \end{aligned}$$

$$\begin{aligned}
&= \Gamma_t(H - H) \\
&= 0,
\end{aligned}$$

where the last steps hold since Γ_t is a constant and by the definition of *valid* occupancy measures.

In addition it holds:

$$\ell_t(x, a) = \Gamma_t + \sum_{i \in [m]} \lambda_{t,i} g_{t,i}(x, a) - r_t(x, a) \geq 1 + \|\lambda_t\|_1 - \sum_{i \in [m]} \lambda_{t,i} - 1 = 0,$$

and similarly,

$$\ell_t(x, a) = \Gamma_t + \sum_{i \in [m]} \lambda_{t,i} g_{t,i}(x, a) - r_t(x, a) \leq \Gamma_t + \sum_{i \in [m]} \lambda_{t,i} = 1 + 2\|\lambda_t\| \leq \Xi_t.$$

This concludes the proof. $\square$

Theorem 2. Under [Condition 2](#), for any $\delta \in (0, 1)$, with probability at least $1 - 11\delta$, it holds:

$$\|\lambda_t\|_1 \leq \Lambda \quad \forall t \in [T + 1],$$

where $\Lambda = \frac{112mH^2}{\rho^2}$.

Proof. Let $M > 1$ be a constant. By absurd suppose $\exists t_2 \in [T]$ s.t.

$$\forall t \leq t_2 \quad \|\lambda_t\|_1 \leq \frac{2HM}{\rho^2} \quad \wedge \quad \|\lambda_{t_2+1}\|_1 > \frac{2HM}{\rho^2} \quad (\text{E.1})$$

and let $t_1 < t_2$ s.t.

$$\|\lambda_{t_1-1}\|_1 \leq \frac{2H}{\rho} \quad \wedge \quad \forall t : t_1 \leq t \leq t_2 \quad \|\lambda_t\|_1 \geq \frac{2H}{\rho}.$$

By construction $1 < \frac{2H}{\rho} \leq \|\lambda_t\|_1 \leq \frac{2HM}{\rho^2}$ for all $t_1 \leq t \leq t_2$, and it holds if $\eta \leq \frac{1}{mH}$:

$$\|\lambda_{t_1}\|_1 \leq \|\lambda_{t_1-1}\|_1 + m\eta H \leq \frac{2H}{\rho} + m\eta H \leq \frac{4H}{\rho}. \quad (\text{E.2})$$

Notice also that by construction, calling $\lambda_{t_1,t_2} = \max_{t \in [t_1, \dots, t_2]} \|\lambda_t\|_1$, it holds:

$$1 < \lambda_{t_1,t_2} \leq \frac{2HM}{\rho^2} \quad \wedge \quad 1 + \lambda_{t_1,t_2} < \frac{4HM}{\rho^2}. \quad (\text{E.3})$$

In the stochastic setting the following holds by Azuma-Hoeffding inequality with probability at least $1 - \delta$:

$$\begin{aligned}
\sum_{t=t_1}^{t_2} -\lambda_t^\top G_t^\top q^\circ &\geq \sum_{t=t_1}^{t_2} -\lambda_t^\top \bar{G}^\top q^\circ - \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^G \\
&\geq \lambda_{t_1,t_2} (t_2 - t_1 + 1) \rho - \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^G \\
&\geq (t_2 - t_1 + 1) 2H - \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^G,
\end{aligned}$$

where $\mathcal{E}_{t_1,t_2}^G = B_1 \sqrt{(t_2 - t_1 + 1)} = 2H \sqrt{\ln(t^2/\delta)} \sqrt{(t_2 - t_1 + 1)}$. Instead, in the adversarial setting, it holds:

$$\begin{aligned}
\sum_{t=t_1}^{t_2} -\lambda_t^\top G_t^\top q^\circ &\geq \sum_{t=t_1}^{t_2} \sum_{i=1}^m -\lambda_{t,i} [G_t^\top q^\circ]_i \\
&\geq \rho \sum_{t=t_1}^{t_2} \sum_{i=1}^m \lambda_{t,i} \\
&= \rho \sum_{t=t_1}^{t_2} \|\lambda_t\|_1 \\
&\geq (t_2 - t_1 + 1) 2H.
\end{aligned}$$

Generalizing the result, it holds, both for the stochastic and the adversarial setting, the following inequality with probability equal to 1 in the adversarial case and with probability at least $1 - \delta$ in the stochastic case:

$$\sum_{t=t_1}^{t_2} -\lambda_t^\top G_t^\top q^\circ \geq (t_2 - t_1 + 1) 2H - \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^G.$$

Thank to this result we can find a lower bound for $-\sum_{t=t_1}^{t_2} \ell_t^{\mathcal{L},\top} q_t$ with probability at least $1 - 9\delta$ in the stochastic setting and with probability at least $1 - 8\delta$ in the adversarial case, employing [Theorem 1](#).

$$-\sum_{t=t_1}^{t_2} \ell_t^{\mathcal{L},\top} q_t = \sum_{t=t_1}^{t_2} (r_t^\top q^\circ - \lambda_t^\top G_t^\top q^\circ) - \sum_{t=t_1}^{t_2} \ell_t^{\mathcal{L},\top} (q_t - q^\circ)$$

$$\geq \sum_{t=t_1}^{t_2} -\lambda^\top G_t^\top q^\circ - \sum_{t=t_1}^{t_2} \ell_t^{\mathcal{L},\top}(q_t - q^\circ) \quad (\text{E.4})$$

$$\geq 2H(t_2 - t_1 + 1) - \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^G - \mathcal{E}_{t_1,t_2}^P, \quad (\text{E.5})$$

where Inequality (E.4) holds since $r_t^\top q^\circ \geq 0$, for all $t \in [T]$, and Inequality (E.5) is derived using the bound on the primal interval regret given by Theorem 1 and defined as $\mathcal{E}_{t_1,t_2}^P$ and by Lemma 2.

At the same time, it is possible to define also an upper bound for the same quantity $-\sum_{t=t_1}^{t_2} \ell_t^{\mathcal{L},\top} q_t$ with probability at least $1 - 2\delta$:

$$\begin{aligned} -\sum_{t=t_1}^{t_2} \ell_t^{\mathcal{L},\top} q_t &= \sum_{t=t_1}^{t_2} (r_t^\top q_t - \lambda_t^\top G_t^\top q_t) \\ &\leq \sum_{t=t_1}^{t_2} H - \sum_{t=t_1}^{t_2} \lambda_t^\top (G_t^\top q_t - \sum_{h=0}^{H-1} G_t(x_h, a_h)) + \sum_{t=t_1}^{t_2} (\underline{0} - \lambda_t) \sum_{h=0}^{H-1} G_t(x_h, a_h) \\ &\leq H(t_2 - t_1 + 1) + \lambda_{t_1,t_2} \sum_{t=t_1}^{t_2} \sum_{x,a} G_t(x, a) (\mathbb{I}_t(x, a) - q_t(x, a)) + \mathcal{E}_{t_1,t_2}^D(\underline{0}) \\ &\leq H(t_2 - t_1 + 1) + \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^{\mathbb{I}} + \mathcal{E}_{t_1,t_2}^D(\underline{0}), \end{aligned}$$

where $\mathcal{E}^{\mathbb{I}} = F_1 \sqrt{(t_2 - t_1 + 1)} = H \sqrt{2 \ln(T^2/\delta)} \sqrt{(t_2 - t_1 + 1)}$ and $\mathcal{E}^D(\underline{0}) = D_1 \frac{\|\lambda_{t_1}\|_2^2}{\eta} + D_2 \eta(t_2 - t_1 + 1) = \frac{1}{2} \frac{\|\lambda_{t_1}\|_2^2}{\eta} + \frac{mH^2}{2} \eta(t_2 - t_1 + 1)$. Thus, combining the two bounds we get with probability at least $1 - 10\delta$ in the adversarial case and $1 - 11\delta$ in the stochastic case the following bound,

$$2H(t_2 - t_1 + 1) - \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^G - \mathcal{E}_{t_1,t_2}^P \leq H(t_2 - t_1 + 1) + \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^{\mathbb{I}} + \mathcal{E}_{t_1,t_2}^D(\underline{0}),$$

which can be reordered as

$$H(t_2 - t_1 + 1) \leq \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^G + \lambda_{t_1,t_2} \mathcal{E}_{t_1,t_2}^{\mathbb{I}} + \mathcal{E}_{t_1,t_2}^D(\underline{0}) + \mathcal{E}_{t_1,t_2}^P.$$

We recall here the definitions of the bounds $\mathcal{E}_{t_1,t_2}^G$, $\mathcal{E}_{t_1,t_2}^{\mathbb{I}}$, $\mathcal{E}_{t_1,t_2}^D(\underline{0})$ and $\mathcal{E}_{t_1,t_2}^P$.

$$\mathcal{E}_{t_1,t_2}^G = B_1 \sqrt{(t_2 - t_1 + 1)},$$

where $B_1 = 2H \sqrt{\ln\left(\frac{T^2}{\delta}\right)}$.

$$\mathcal{E}_{t_1,t_2}^{\mathbb{I}} = F_1 \sqrt{(t_2 - t_1 + 1)},$$

where $F_1 = H \sqrt{2 \ln\left(\frac{T^2}{\delta}\right)}$.

$$\mathcal{E}_{t_1,t_2}^D(\underline{0}) = D_1 \frac{\|\lambda_{t_1}\|_2^2}{\eta} + D_2 \eta(t_2 - t_1 + 1),$$

where $D_1 = \frac{1}{2}$, $D_2 = \frac{mH^2}{2}$.

$$\mathcal{E}_{t_1,t_2}^P = U_1 \Xi_{t_1,t_2} C \sqrt{T} + U_2 \Xi_{t_1,t_2} \frac{(t_2 - t_1 + 1)}{C \sqrt{T}} + U_3 \Xi_{t_1,t_2} \frac{1}{C \sqrt{T}} + U_4 \Xi_{t_1,t_2} \sqrt{T},$$

where $U_1 = 6H^2 \ln\left(\frac{H|A|T^2}{\delta}\right)$, $U_2 = 9H|X||A|$, $U_3 = \frac{H}{2} \ln\left(\frac{HT^2}{\delta}\right)$ and $U_4 = 30H^2|X|^2 \sqrt{2|A| \ln\left(\frac{T|X|^2|A|}{\delta}\right)}$.

Thus, we can write:

$$\begin{aligned} H(t_2 - t_1 + 1) &\leq \underbrace{\lambda_{t_1,t_2} (F_1 + B_1) \sqrt{(t_2 - t_1 + 1)}}_{\textcircled{1}} + \underbrace{U_2 \Xi_{t_1,t_2} \frac{t_2 - t_1 + 1}{C \sqrt{T}}}_{\textcircled{2}} \\ &\quad + \underbrace{U_3 \Xi_{t_1,t_2} \frac{1}{C \sqrt{T}}}_{\textcircled{3}} + \underbrace{U_1 \Xi_{t_1,t_2} C \sqrt{T}}_{\textcircled{4}} + \underbrace{D_2 \eta(t_2 - t_1 + 1)}_{\textcircled{5}} \\ &\quad + \underbrace{D_1 \frac{\|\lambda_{t_1}\|_2^2}{\eta}}_{\textcircled{6}} + \underbrace{U_4 \Xi_{t_1,t_2} \sqrt{T}}_{\textcircled{7}}. \end{aligned}$$

To conclude the proof by absurd it is sufficient to prove that all ①, ②, ③, ④, ⑤, ⑥, ⑦ are smaller or equal to $\frac{H(t_2-t_1+1)}{7}$, with at least one being strictly smaller.

Prove ① $< \frac{H(t_2-t_1+1)}{7}$ If $\eta \leq \frac{1}{14m(F_1+B_1)\sqrt{T}}$, then ① $< \frac{H(t_2-t_1+1)}{7}$ holds. Indeed:

$$\frac{H(t_2-t_1+1)}{7} > \frac{HM}{7\rho^2 m\eta} \quad (\text{E.6a})$$

$$\geq \frac{2HM}{\rho^2} (F_1+B_1)\sqrt{T} \quad (\text{E.6b})$$

$$\geq \lambda_{t_1,t_2} (F_1+B_1)\sqrt{T} \quad (\text{E.6c})$$

$$\geq \lambda_{t_1,t_2} (F_1+B_1)\sqrt{t_2-t_1+1},$$

where Inequality (E.6a) holds by Lemma 11, Inequality (E.6b) is equivalent to condition $\eta \leq \frac{1}{14m(F_1+B_1)\sqrt{T}}$ and Inequality (E.6c) is true by Assumption (E.1).

Prove ② $< \frac{H(t_2-t_1+1)}{7}$ If $C \geq 56 \frac{MU_2}{\rho^2\sqrt{T}}$ holds, then ② $< \frac{H(t_2-t_1+1)}{7}$ also holds. Indeed:

$$\frac{H(t_2-t_1+1)}{7} \geq 2U_2 \left(\frac{4HM}{\rho^2} \right) \frac{t_2-t_1+1}{C\sqrt{T}} \quad (\text{E.7a})$$

$$> U_2 2(1+\lambda_{t_1,t_2}) \frac{t_2-t_1+1}{C\sqrt{T}} \quad (\text{E.7b})$$

$$\geq U_2 \Xi_{t_1,t_2} \frac{t_2-t_1+1}{C\sqrt{T}}, \quad (\text{E.7c})$$

where Inequality (E.7a) is equivalent to the condition $C \geq 56 \frac{MU_2}{\rho^2\sqrt{T}}$, Inequality (E.7b) holds by Inequality (E.3) and Inequality (E.7c) is true since $\Xi_{t_1,t_2} \leq 2(1+\lambda_{t_1,t_2})$.

Prove ③ $< \frac{H(t_2-t_1+1)}{7}$ If $\eta \leq \frac{C\sqrt{T}}{56mU_3}$ holds then also ③ $< \frac{H(t_2-t_1+1)}{7}$ holds. Indeed:

$$\frac{H(t_2-t_1+1)}{7} > \frac{HM}{7\rho^2 m\eta} \quad (\text{E.8a})$$

$$\geq U_3 2 \frac{4HM}{\rho^2} \frac{1}{C\sqrt{T}} \quad (\text{E.8b})$$

$$\geq U_3 2(1+\lambda_{t_1,t_2}) \frac{1}{C\sqrt{T}} \quad (\text{E.8c})$$

$$\geq U_3 \Xi_{t_1,t_2} \frac{1}{C\sqrt{T}}, \quad (\text{E.8d})$$

where Inequality (E.8a) hold by Lemma 11, Inequality (E.8b) holds if condition $\eta \leq \frac{C\sqrt{T}}{56mU_3}$ holds, and Inequality (E.8c) and Inequality (E.8d) follow the same reasoning as Inequality (E.7b) and Inequality (E.7c).

Prove ④ $< \frac{H(t_2-t_1+1)}{7}$ If $\eta \leq \frac{1}{56mU_1 C\sqrt{T}}$ holds then also ④ $< \frac{H(t_2-t_1+1)}{7}$ holds. Indeed:

$$\frac{H(t_2-t_1+1)}{7} > \frac{HM}{7\rho^2 m\eta} \quad (\text{E.9})$$

$$\geq U_1 2 \frac{4HM}{\rho^2} C\sqrt{T}$$

$$\geq U_1 2(1+\lambda_{t_1,t_2}) C\sqrt{T}$$

$$\geq U_1 \Xi_{t_1,t_2} C\sqrt{T},$$

where Inequality (E.9) holds when condition $\eta \leq \frac{1}{56mU_1 C\sqrt{T}}$ also holds, and the rest of the inequalities follow a similar reasoning to the one used to bound ③.

Prove ⑤ $\leq \frac{H(t_2-t_1+1)}{7}$ It is immediate to see that if $\eta \leq \frac{H}{7D_2}$ holds, then it holds also that:

$$\textcircled{5} = D_2 \eta (t_2 - t_1 + 1) \leq \frac{H(t_2 - t_1 + 1)}{7}.$$

Prove ⑥ $< \frac{H(t_2-t_1+1)}{7}$

If the condition $M \geq 112D_1 H m$ is satisfied than the inequality $\textcircled{6} < \frac{H(t_2-t_1+1)}{7}$ holds too. Indeed:

$$\frac{H(t_2-t_1+1)}{7} > \frac{HM}{7\rho^2 m \eta} \quad (\text{E.10a})$$

$$\geq D_1 \frac{16H^2}{\rho^2} \frac{1}{\eta} \quad (\text{E.10b})$$

$$\geq D_1 \frac{\|\lambda_{t_1}\|_1^2}{\eta} \quad (\text{E.10c})$$

$$\geq D_1 \frac{\|\lambda_{t_1}\|_2^2}{\eta},$$

where Inequality E.10a holds by Lemma 11, Inequality (E.10b) holds when the condition $M \geq 112D_1 H m$ is satisfied and Inequality (E.10c) holds by Inequality (E.2).

Prove $\textcircled{7} < \frac{H(t_2-t_1+1)}{7}$

If the condition $\eta \leq \frac{1}{56mU_4\sqrt{T}}$ is satisfied then $\textcircled{7} < \frac{H(t_2-t_1+1)}{7}$ also holds. In fact

$$\frac{H(t_2-t_1+1)}{7} > \frac{HM}{7\rho^2 m \eta} \quad (\text{E.11a})$$

$$\geq U_4 2 \frac{4HM}{\rho^2} \sqrt{T} \quad (\text{E.11b})$$

$$\geq U_4 2(1 + \lambda_{t_1, t_2}) \sqrt{T}$$

$$\geq U_4 \Xi_{t_1, t_2} \sqrt{T}.$$

where Inequality (E.11a) holds by Lemma 11 and inequality (E.11b) holds if condition $\eta \leq \frac{1}{56mU_4\sqrt{T}}$ also holds.

Conclusion of the proof. Thus, we have the following 3 conditions:

- First condition:

$$\begin{aligned} M &\geq 112D_1 H m \\ &= 112 \frac{1}{2} H m \\ &= 56 H m. \end{aligned}$$

- Second condition:

$$\begin{aligned} C &\geq 56 \frac{MU_2}{\rho^2 \sqrt{T}} \\ &= 56 \frac{M9H|X||A|}{\rho^2 \sqrt{T}}. \end{aligned}$$

- Third condition:

$$\eta \leq \min \left\{ \frac{1}{14m(F_1 + B_1)\sqrt{T}}, \frac{C\sqrt{T}}{56mU_3}, \frac{1}{56mU_1 C \sqrt{T}}, \frac{H}{7D_2}, \frac{1}{56mU_4 \sqrt{T}} \right\}.$$

Thus, we set M as $M = 56Hm$, and consequently, under Condition 2 we set $C = 252|X||A|H$ since

$$\begin{aligned} C &= 252|X||A|H \\ &\geq 252|X||A| \\ &\geq 252|X||A| \frac{112mH^2}{\rho^2} \frac{1}{\sqrt{T}} \\ &= 56 \frac{(56Hm)9H|X||A|}{\rho^2 \sqrt{T}} \\ &= 56 \frac{9MH|X||A|}{\rho^2 \sqrt{T}}. \end{aligned}$$

Notice that the inequality is deduced directly by Condition 2. In fact if $\rho \geq T^{-\frac{1}{8}} H \sqrt{112m}$ then it is also true that

$$\frac{112mH^2}{\rho^2} \leq T^{\frac{1}{4}} \leq \sqrt{T}.$$

As a final remark, we choose $252|X||A|H$ as value of C instead of the smaller value $252|X||A|$, which is useful for [Lemma 6](#). Finally we study the condition on η .

$$\begin{aligned} & \min \left\{ \frac{1}{14m(F_1 + B_1)\sqrt{T}}, \frac{C\sqrt{T}}{56mU_3}, \frac{1}{56mU_1C\sqrt{T}}, \frac{H}{7D_2}, \frac{1}{56mU_4\sqrt{T}} \right\} \\ & \geq \min \left\{ \frac{1}{14m \left(4H \sqrt{\ln \left(\frac{T^2}{\delta} \right)} \right) \sqrt{T}}, \frac{252|X||A|H\sqrt{T}}{56m \left(\frac{H}{2} \ln \left(\frac{HT^2}{\delta} \right) \right)}, \right. \\ & \quad \frac{1}{56m \left(6H^2 \ln \left(\frac{H|A|T^2}{\delta} \right) \right) (252|X||A|H)\sqrt{T}}, \frac{H}{7 \left(\frac{mH^2}{2} \right)}, \\ & \quad \left. \frac{1}{56m \left(30H^2|X|^2 \sqrt{2|A| \ln \left(\frac{T|X|^2|A|}{\delta} \right)} \right) \sqrt{T}} \right\} \\ & \geq \frac{1}{84672mH^2|X|^2|A| \ln \left(\frac{|A||X|^2T^2}{\delta} \right) \sqrt{T}}. \end{aligned}$$

Thus, the proof is concluded taking $\eta = \frac{1}{84672mH^2|X|^2|A| \ln \left(\frac{|A||X|^2T^2}{\delta} \right) \sqrt{T}}$. $\square$

Lemma 3. If [Condition 2](#) holds, for all $t \in [T]$ and for each constraints $i \in [m]$, it holds:

$$\lambda_{t,i} \geq \eta \hat{V}_{t-1,i},$$

where $\hat{V}_{t,i} := \sum_{\tau=1}^t \sum_{x,a} g_{\tau,i}(x, a) \mathbb{I}_{\tau}(x, a)$.

Proof. First observe that with $t = 1$ we have that $\hat{V}_{t-1,i}$ is the sum of zero elements and as such, it is equal to zero. This means that for $t = 1$ the inequality $\lambda_{t,i} \geq \eta \hat{V}_{t-1,i}$ is equivalent to

$$\lambda_{t,i} \geq 0,$$

which is true by construction. We finish the proof by induction. Suppose $\lambda_{t,i} \geq \eta \hat{V}_{t-1,i}$ is true for a $t \in [T]$, we show that it also holds for $t + 1$, indeed:

$$\begin{aligned} \lambda_{t+1,i} &= \max \left\{ \lambda_{t,i} + \eta \sum_{h=0}^{H-1} g_{t,i}(x_h, a_h), 0 \right\} \\ &= \max \left\{ \lambda_{t,i} + \eta \sum_{x,a} g_{t,i}(x, a) \mathbb{I}_t(x, a), 0 \right\} \\ &\geq \lambda_{t,i} + \eta \sum_{x,a} g_{t,i}(x, a) \mathbb{I}_t(x, a) \\ &\geq \eta \hat{V}_{t-1,i} + \eta \sum_{x,a} g_{t,i}(x, a) \mathbb{I}_t(x, a) \\ &= \eta \left(\sum_{\tau=1}^{t-1} g_{\tau,i}(x, a) \mathbb{I}_{\tau}(x, a) + g_{t,i}(x, a) \mathbb{I}_t(x, a) \right) \\ &= \eta \sum_{\tau=1}^t g_{\tau,i}(x, a) \mathbb{I}_{\tau}(x, a) \\ &= \eta \hat{V}_{t,i}. \end{aligned}$$

This concludes the proof. $\square$

Lemma 4. If [Condition 2](#) holds, referring as i^* to the element in $[m]$ such that $i^* = \arg \max_{i \in [m]} \sum_{t=1}^T [G_t^\top q^{P, \pi_t}]_i$, then with probability at least $1 - \delta$, it holds:

$$V_T \leq \hat{V}_{T,i^*} + \mathcal{E}^\dagger.$$

Proof. We observe with probability at least $1 - \delta$:

$$\begin{aligned} V_T &= \sum_{t=1}^T [G_t^\top q^{P, \pi_t}]_{i^*} \\ &= \sum_{t=1}^T \sum_{x,a} g_{t,i^*}(x, a) (q_t(x, a) - \mathbb{I}_t(x, a)) + \sum_{t=1}^T \sum_{x,a} g_{t,i^*}(x, a) \mathbb{I}_t(x, a) \end{aligned}$$

$$\begin{aligned} &\leq \sum_{t=1}^T \sum_{x,a} g_{t,i^*}(x,a) (q_t(x,a) - \mathbb{I}_t(x,a)) + \widehat{V}_{T,i^*} \\ &\leq \mathcal{E}^\dagger + V_{T,i^*}. \end{aligned}$$

This concludes the proof. $\square$

Lemma 5. When [Condition 2](#) does not hold, with probability at least $1 - 10\delta$ in case of stochastic costs and $1 - 9\delta$ in case of adversarial costs it holds for all $i \in [m]$:

$$\widehat{V}_{T,i} \leq \frac{4T^{\frac{1}{4}}}{\eta}.$$

Proof. Recall the definition of $\widehat{V}_{T,i}$ as $\widehat{V}_{T,i} = \sum_{t=1}^T \sum_{x,a} g_{t,i}(x,a) \mathbb{I}_t(x,a)$. We first focus on the stochastic setting. Thus, with probability at least $1 - \delta$, it holds:

$$\begin{aligned} \sum_{t=1}^T r_t^\top q_t - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t &= \sum_{t=1}^T r_t^\top q_t^\circ - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t^\circ + \sum_{t=1}^T \ell_t^{\mathcal{L},\top} (q_t^\circ - q_t) \\ &\geq - \sum_{t=1}^T \lambda_t^\top \overline{G}^\top q_t^\circ - \lambda_{1,T} \mathcal{E}_T^G + \sum_{t=1}^T \ell_t^{\mathcal{L},\top} (q_t^\circ - q_t) \\ &\geq -mT^{\frac{1}{4}} \mathcal{E}_T^G + \sum_{t=1}^T \ell_t^{\mathcal{L},\top} (q_t^\circ - q_t). \end{aligned}$$

On the other hand, in case of adversarial constraints, it holds:

$$\begin{aligned} \sum_{t=1}^T r_t^\top q_t - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t &= \sum_{t=1}^T r_t^\top q_t^\circ - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t^\circ + \sum_{t=1}^T \ell_t^{\mathcal{L},\top} (q_t^\circ - q_t) \\ &\geq \sum_{t=1}^T \ell_t^{\mathcal{L},\top} (q_t^\circ - q_t). \end{aligned}$$

Define a vector $\tilde{\lambda} \in [0, T^{\frac{1}{4}}]^m$ as $\tilde{\lambda}_j = 0$ if $j \neq i$ and $\tilde{\lambda}_j = T^{\frac{1}{4}}$ if $j = i$. Simultaneously with probability at least $1 - \delta$ it holds:

$$\begin{aligned} &\sum_{t=1}^T r_t^\top q_t - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t \\ &\leq \sum_{t=1}^T r_t^\top q_t - \sum_{t=1}^T \tilde{\lambda}^\top \sum_{x,a} G_t(x,a) \mathbb{I}_t(x,a) + \sum_{t=1}^T (\tilde{\lambda} - \lambda_t)^\top \sum_{x,a} G_t(x,a) \mathbb{I}_t(x,a) + \lambda_{1,T} \mathcal{E}^\dagger \\ &\leq \sum_{t=1}^T r_t^\top q_t - \sum_{t=1}^T \tilde{\lambda}^\top \sum_{x,a} G_t(x,a) \mathbb{I}_t(x,a) + \mathcal{E}_T^D(\tilde{\lambda}) + mT^{\frac{1}{4}} \mathcal{E}^\dagger \\ &\leq HT - T^{\frac{1}{4}} \widehat{V}_{T,i} + \mathcal{E}_T^D(\tilde{\lambda}) + mT^{\frac{1}{4}} \mathcal{E}^\dagger, \end{aligned}$$

where in the first equality we used the definition of $\mathcal{E}_T^\dagger$, in the first inequality we used the definition of the dual space $[0, T^{\frac{1}{4}}]^m$ to bound $\lambda_{1,T}$ as $mT^{\frac{1}{4}}$, and in the last inequality we used the definition of $\tilde{\lambda}$. We can then compare the lower and the upper bound for $\sum_{t=1}^T r_t^\top q_t - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t$ obtaining the following inequality, which holds with probability at least $1 - \delta$ with adversarial constraints and with probability at least $1 - 2\delta$ with stochastic constraints:

$$-mT^{\frac{1}{4}} \mathcal{E}_T^G + \sum_{t=1}^T \ell_t^{\mathcal{L},\top} (q_t^\circ - q_t) \leq HT - T^{\frac{1}{4}} \widehat{V}_{T,i} + \mathcal{E}_T^D(\tilde{\lambda}) + mT^{\frac{1}{4}} \mathcal{E}^\dagger,$$

from which we can write the following inequality that holds with probability at least $1 - 9\delta$ with adversarial constraints and $1 - 10\delta$ with stochastic constraints:

$$T^{\frac{1}{4}} \widehat{V}_{T,i} \leq mT^{\frac{1}{4}} (\mathcal{E}_T^G + \mathcal{E}_T^\dagger) + \mathcal{E}_T^P + \mathcal{E}_T^D(\tilde{\lambda}) + HT. \quad (\text{E.12})$$

We proceed now to bound each element of the right side of the inequality.

To bound $\mathcal{E}^P$ we use the fact that $\Xi_{1,T} \leq (1 + \lambda_{1,T}) \leq (1 + mT^{\frac{1}{4}})$ and the definition of η as following:

$$\begin{aligned} \mathcal{E}_T^P &= \Xi_{1,T} \left(U_1 C \sqrt{T} + U_2 \frac{\sqrt{T}}{C} + U_3 \frac{1}{C \sqrt{T}} \right) + U_4 \sqrt{T} \\ &\leq 2(1 + mT^{\frac{1}{4}}) \sqrt{T} \left(1512 H^3 |X| |A| \ln \left(\frac{H|A|T^2}{\delta} \right) + \frac{9H|X||A|}{252H|X||A|} + \frac{\frac{H}{2} \ln \left(\frac{HT^2}{\delta} \right)}{252H|X||A|} \right) \end{aligned}$$

$$\begin{aligned}
& + \sqrt{T}30H^2|X|^2 \sqrt{2|A| \ln \left(\frac{T|X|^2|A|}{\delta} \right)} \\
& \leq T^{\frac{1}{4}} \sqrt{T}6056H^3m|X||A| \ln \left(\frac{H|A|T^2}{\delta} \right) + \sqrt{T}30H^2|X|^2 \sqrt{2|A| \ln \left(\frac{T|X|^2|A|}{\delta} \right)} \\
& \leq T^{\frac{1}{4}} \sqrt{T}6116H^2m|X|^2|A| \ln \left(\frac{|X|^2|A|T^2}{\delta} \right) \\
& \leq \frac{T^{\frac{1}{4}}}{\eta}
\end{aligned}$$

To bound $\mathcal{E}_T^D(\tilde{\lambda})$ we use [Theorem 11](#), the fact that by its definition $\|\tilde{\lambda}\|_2^2 = \left(T^{\frac{1}{4}}\right)^2 = \sqrt{T}$, the initialization of the dual $\lambda_1 = \underline{0}$ and the definition of η in the following way:

$$\begin{aligned}
\mathcal{E}_T^D(\tilde{\lambda}) & \leq \frac{\|\lambda_1 - \tilde{\lambda}\|_2^2}{2\eta} + \frac{\eta}{2}TmH^2 \\
& = \frac{\|\tilde{\lambda}\|_2^2}{2\eta} + \frac{\eta}{2}TmH^2 \\
& = \frac{\sqrt{T}}{2\eta} + \frac{\eta}{2}TmH^2 \\
& \leq \frac{\sqrt{T}}{2\eta} + \frac{mH^2T}{2} \frac{1}{84672mH^2|X|^2|A| \ln \left(\frac{|A||X|^2T^2}{\delta} \right) \sqrt{T}} \\
& \leq \frac{\sqrt{T}}{2\eta} + \frac{\sqrt{T}}{2\eta} = \frac{\sqrt{T}}{\eta}
\end{aligned}$$

We proceed to simply bound also $mT^{\frac{1}{4}}(\mathcal{E}_T^G + \mathcal{E}_T^I)$ through their definition:

$$\begin{aligned}
mT^{\frac{1}{4}}(\mathcal{E}_T^G + \mathcal{E}_T^I) & = mT^{\frac{1}{4}} \sqrt{T} \left(2H \sqrt{\ln \left(\frac{T^2}{\delta} \right)} + H \sqrt{2 \ln \left(\frac{T^2}{\delta} \right)} \right) \\
& \leq \frac{T^{\frac{1}{4}}}{\eta}
\end{aligned}$$

Finally we bound HT as $HT \leq \frac{\sqrt{T}}{\eta}$.

Thus, Inequality (E.12) becomes

$$\hat{V}_{T,i} \leq \frac{1}{T^{\frac{1}{4}}} \left(mT^{\frac{1}{4}}(\mathcal{E}_T^G + \mathcal{E}_T^I) + \mathcal{E}_T^P + \mathcal{E}_T^D(\tilde{\lambda}) + HT \right) \leq \frac{4\sqrt{T}}{T^{\frac{1}{4}}\eta} = \frac{4T^{\frac{1}{4}}}{\eta},$$

which concludes the proof. $\square$

Appendix F. Analysis with stochastic constraints

Theorem 6. Suppose that [Condition 2](#) holds and the constraints are generated stochastically. Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{\mathcal{O}}(\Lambda \sqrt{T}), \quad V_T \leq \tilde{\mathcal{O}}(\Lambda \sqrt{T}),$$

with probability at least $1 - 14\delta$ when the rewards are stochastic, and with probability at least $1 - 13\delta$ when the rewards are adversarial.

Proof. With probability at least $1 - 12\delta$ it holds:

$$V_T \leq \hat{V}_{T,i^*} + \mathcal{E}^I \tag{F.1a}$$

$$\leq \frac{1}{\eta} \lambda_{T+1,i^*} + \mathcal{E}^I \tag{F.1b}$$

$$\leq \frac{1}{\eta} \Lambda + \mathcal{E}^I, \tag{F.1c}$$

where Inequality (F.1a) holds by [Lemma 4](#), Inequality (F.1b) holds by [Lemma 3](#) and Inequality (F.1c) holds by [Theorem 2](#). Then, with probability at least $1 - 12\delta$ we observe that:

$$\sum_{t=1}^T r_t^\top q^* - \sum_{t=1}^T r_t^\top q^{P,x_t} \leq \sum_{t=1}^T (r_t^\top q^* - \lambda_t^\top G_t^\top q^*) - \sum_{t=1}^T (r_t^\top q_t - \lambda_t^\top G_t^\top q_t) + \sum_{t=1}^T \lambda_t^\top G_t^\top (q^* - q_t)$$

$$\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^{\mathbb{I}} + \sum_{t=1}^T \lambda_t^\top G_t^\top q^* \quad (\text{F.2a})$$

$$\begin{aligned} &= \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^{\mathbb{I}} + \sum_{t=1}^T \lambda_t^\top (G_t - \bar{G})^\top q^* + \sum_{t=1}^T \lambda_t^\top \bar{G}^\top q^* \\ &\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^{\mathbb{I}} + \lambda_{1,T} \mathcal{E}^G \quad (\text{F.2b}) \\ &\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \Lambda \mathcal{E}^{\mathbb{I}} + \Lambda \mathcal{E}^G, \quad (\text{F.2c}) \end{aligned}$$

where Inequality (F.2a) holds by Theorem 1 and by Theorem 11, Inequality (F.2b) holds since in the stochastic constraint case $\sum_{t=1}^T (G_t - \bar{G})^\top q^* \leq \mathcal{E}^G$ with probability at least $1 - \delta$ by definition of $\mathcal{E}^G$, and finally Inequality (F.2c) holds by Theorem 2. Finally we observe that in the stochastic case with probability at least $1 - \delta$:

$$\left(T \cdot \text{OPT}_{\bar{r}, \bar{G}} - \sum_{t=1}^T r_t^\top q_t \right) - \sum_{t=1}^T r_t^\top (q^* - q_t) \leq \mathcal{E}^r.$$

Thus, if the rewards are stochastic with probability at least $1 - 14\delta$ it holds:

$$R_T \leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \Lambda \mathcal{E}^{\mathbb{I}} + \Lambda \mathcal{E}^G + \mathcal{E}^r, \quad V_T \leq \frac{1}{\eta} \Lambda + \mathcal{E}^{\mathbb{I}}$$

and if the rewards are adversarial with probability at least $1 - 13\delta$ it holds:

$$R_T \leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \Lambda \mathcal{E}^{\mathbb{I}} + \Lambda \mathcal{E}^G, \quad V_T \leq \frac{1}{\eta} \Lambda + \mathcal{E}^{\mathbb{I}}$$

which concludes the proof. $\square$

Theorem 4. Suppose that Condition 2 does not hold and the constraints are generated stochastically. Then, for any $\delta \in (0, 1)$, Algorithm 2 attains:

$$R_T \leq \tilde{\mathcal{O}}(T^{3/4}), \quad V_T \leq \tilde{\mathcal{O}}(T^{3/4}),$$

with probability at least $1 - 11\delta$ when the rewards are stochastic, and with probability at least $1 - 10\delta$ when the rewards are adversarial.

Proof. Similar to the proof of Lemma 3 it holds with probability at least $1 - 10\delta$:

$$\begin{aligned} \sum_{t=1}^T r_t^\top q^* - \sum_{t=1}^T r_t^\top q^{\pi_t} &\leq \sum_{t=1}^T (r_t^\top q^* - \lambda_t^\top G_t^\top q^*) - \sum_{t=1}^T (r_t^\top q_t - \lambda_t^\top G_t^\top q_t) + \sum_{t=1}^T \lambda_t^\top G_t^\top (q^* - q_t) \\ &\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^{\mathbb{I}} + \sum_{t=1}^T \lambda_t^\top G_t^\top q^* \\ &= \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^{\mathbb{I}} + \sum_{t=1}^T \lambda_t^\top (G_t - \bar{G})^\top q^* + \sum_{t=1}^T \lambda_t^\top \bar{G}^\top q^* \\ &\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^{\mathbb{I}} + \lambda_{1,T} \mathcal{E}^G, \end{aligned}$$

therefore with probability at least $1 - 10\delta$ following the reasoning of Lemma 5 it holds with adversarial rewards:

$$\sum_{t=1}^T r_t^\top q_t \geq T \cdot \text{OPT}_{\bar{r}, \bar{G}} - mT^{1/4} \mathcal{E}^G + mT^{1/4} \mathcal{E}^{\mathbb{I}} - \mathcal{E}^D(\underline{0}) - \mathcal{E}^P,$$

and with stochastic rewards with probability at least $1 - 11\delta$:

$$\sum_{t=1}^T r_t^\top q_t \geq T \cdot \text{OPT}_{\bar{r}, \bar{G}} - mT^{1/4} \mathcal{E}^G + mT^{1/4} \mathcal{E}^{\mathbb{I}} - \mathcal{E}^D(\underline{0}) - \mathcal{E}^P - \mathcal{E}^r.$$

Applying Lemma 5 to bound the constraints violation concludes the proof. $\square$

Appendix G. Analysis with adversarial constraints

Theorem 5. Suppose Condition 2 holds and the constraints are adversarial. Then, for any $\delta \in (0, 1)$, Algorithm 2 attains:

$$\sum_{t=1}^T r_t^\top q_t \geq \Omega\left(\frac{\rho}{\rho + H} \cdot \text{OPT}_{\bar{r}, \bar{G}}\right), \quad V_T \leq \tilde{\mathcal{O}}(\Lambda \sqrt{T}),$$

with probability at least $1 - 14\delta$ when the rewards are stochastic, and with probability at least $1 - 13\delta$ when the rewards are adversarial.

Proof.

Thanks to [Theorem 1](#), [Theorems 11](#) and [2](#) with probability at least $1 - 11\delta$ it holds for all $q \in \Delta(P)$:

$$\begin{aligned}
& \sum_{t=1}^T r_t^\top q - \sum_{t=1}^T r_t^\top q_t \\
& \leq - \sum_{t=1}^T \ell_t^{\mathcal{L}, \top} q + \sum_{t=1}^T \ell_t^{\mathcal{L}, \top} q_t + \sum_{t=1}^T \lambda_t^\top G_t^\top q - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t \\
& \leq \mathcal{E}_T^P + \sum_{t=1}^T \lambda_t^\top G_t^\top q + \sum_{t=1}^T (\underline{0} - \lambda_t)^\top \sum_{h=0}^{H-1} G_t(x_h, a_h) + \sum_{t=1}^T \lambda_t^\top \left(\sum_{h=0}^{H-1} G_t(x_h, a_h) - G_t^\top q_t \right) \\
& \leq \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^\natural + \sum_{t=1}^T \sum_{i=1}^m \lambda_{t,i} g_{t,i}^\top q \\
& \leq \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^\natural + \sum_{t=1}^T \sum_{i=1}^m \lambda_{t,i} g_{t,i}^\top q \\
& \leq \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + \Lambda \mathcal{E}^\natural + \sum_{t=1}^T \sum_{i=1}^m \lambda_{t,i} g_{t,i}^\top q.
\end{aligned}$$

Consider now the occupancy measure $\tilde{q} = \frac{\rho}{H+\rho} q^* + \frac{H}{H+\rho} q^\circ$. For all $i \in [m]$ and for all $t \in [T]$:

$$\begin{aligned}
g_{t,i}^\top \tilde{q} & \leq \left(\frac{\rho}{H+\rho} g_{t,i}^\top q^* + \frac{H}{H+\rho} g_{t,i}^\top q^\circ \right) \\
& \leq \left(\frac{H\rho}{H+\rho} - \frac{H\rho}{H+\rho} \right) \\
& = 0,
\end{aligned}$$

given that $g_{t,i}^\top q^* \leq \|q^*\|_1 \leq H$, and $g_{t,i}^\top q^\circ \leq -\rho$ by definition of q° and by definition of ρ .

$$\begin{aligned}
\sum_{t=1}^T r_t^\top \tilde{q} & = \sum_{t=1}^T \left(\frac{\rho}{H+\rho} r_t^\top q^* + \frac{H}{H+\rho} r_t^\top q^\circ \right) \\
& \geq \frac{\rho}{H+\rho} \sum_{t=1}^T r_t^\top q^*,
\end{aligned}$$

since $r_t^\top q^\circ \geq 0$. Notice also that with adversarial rewards $\sum_{t=1}^T r_t^\top q^* = T \cdot \text{OPT}_{\bar{r}, \bar{G}}$, while with stochastic rewards with probability at least $1 - \delta$ it holds $\sum_{t=1}^T r_t^\top q^* \geq T \cdot \text{OPT}_{\bar{r}, \bar{G}} - \mathcal{E}^r$, by definition of $\mathcal{E}^r$ and $\text{OPT}_{\bar{r}, \bar{G}}$ for stochastic rewards. By reordering the terms we get that with probability at least $1 - 11\delta$

$$\sum_{t=1}^T r_t^\top q_t \geq \frac{\rho}{H+\rho} \sum_{t=1}^T r_t^\top q^* - \mathcal{E}_T^P - \mathcal{E}_T^D(\underline{0}) - \Lambda \mathcal{E}^\natural,$$

we can proceed to bound the regret in both cases: adversarial rewards and stochastic rewards.

With probability at least $1 - 11\delta$ with adversarial rewards it holds:

$$\begin{aligned}
R_T & = \sum_{t=1}^T r_t^\top q^* - \sum_{t=1}^T r_t^\top q_t \\
& \leq \sum_{t=1}^T r_t^\top q^* - \left(\frac{\rho}{H+\rho} \sum_{t=1}^T r_t^\top q^* - \mathcal{E}_T^P - \mathcal{E}_T^D(\underline{0}) - \Lambda \mathcal{E}^\natural \right) \\
& \leq \frac{H}{H+\rho} \sum_{t=1}^T r_t^\top q^* + \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + \Lambda \mathcal{E}^\natural \\
& \leq \frac{H}{H+\rho} T \cdot \text{OPT}_{\bar{r}, \bar{G}} + \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + \Lambda \mathcal{E}^\natural.
\end{aligned}$$

With stochastic rewards it holds with probability at least $1 - 11\delta$:

$$\sum_{t=1}^T r_t^\top q_t \geq \frac{\rho}{H+\rho} \sum_{t=1}^T r_t^\top q^* - \mathcal{E}_T^P - \mathcal{E}_T^D(\underline{0}) - \Lambda \mathcal{E}^\natural,$$

and with probability at least $1 - 12\delta$:

$$\sum_{t=1}^T r_t^\top q_t \geq \frac{\rho}{H+\rho} T \cdot \text{OPT}_{\bar{r}, \bar{G}} - \mathcal{E}_T^P - \mathcal{E}_T^D(\underline{0}) - \Lambda \mathcal{E}^\natural - \mathcal{E}^r.$$

To conclude the proof we observe that following the analogous reasoning to [Theorem 6](#) in case of adversarial constraints it also holds with probability at least $1 - 12\delta$:

$$V_T \leq \frac{1}{\eta} \Lambda + \mathcal{E}^\dagger.$$

□

Appendix H. Analysis with limited adversariality

In the following, we will refer as $\hat{G}$ to the constraint matrix that minimize the definition of B , that is $\hat{G} := \arg \min_{G \in [-1, 1]^{|X \times A| \times m}} \sum_{t=1}^T \|G_t - G\|_1$.

Theorem 6. Suppose that [Condition 2](#) holds and the constraints are generated adversarially and are parameterized given B . Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains the following bounds:

$$R_T \leq \tilde{\mathcal{O}}(\Lambda(\sqrt{T} + B)), \quad V_T \leq \tilde{\mathcal{O}}(\Lambda\sqrt{T}),$$

with probability at least $1 - 14\delta$ when the rewards are stochastic, and with probability at least $1 - 13\delta$ when the rewards are adversarial.

Proof. Analogously to [Theorem 6](#) it holds with probability at least $1 - 12\delta$:

$$V_T \leq \frac{1}{\eta} \Lambda + \mathcal{E}^\dagger,$$

Then, with probability at least $1 - 12\delta$ we observe that:

$$\begin{aligned} \sum_{t=1}^T r_t^\top q^* - \sum_{t=1}^T r_t^\top q^{P, \pi_t} &\leq \sum_{t=1}^T (r_t^\top q^* - \lambda_t^\top G_t^\top q^*) - \sum_{t=1}^T (r_t^\top q_t - \lambda_t^\top G_t^\top q_t) + \sum_{t=1}^T \lambda_t^\top G_t^\top (q^* - q_t) \\ &\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^\dagger + \sum_{t=1}^T \lambda_t^\top G_t^\top q^* \end{aligned} \quad (\text{H.1a})$$

$$\begin{aligned} &= \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^\dagger + \sum_{t=1}^T \lambda_t^\top (G_t - \bar{G})^\top q^* + \sum_{t=1}^T \lambda_t^\top \bar{G}^\top q^* \\ &\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^\dagger + 2\lambda_{1,T} B \end{aligned} \quad (\text{H.1b})$$

$$\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \Lambda \mathcal{E}^\dagger + 2\Lambda B, \quad (\text{H.1c})$$

where Inequality [\(H.1a\)](#) holds by [Theorem 1](#) and by [Theorem 11](#), Inequality [\(H.1b\)](#) holds since in the non-stationary constraint case $\sum_{t=1}^T (G_t - \bar{G})^\top q^* \leq \sum_{t=1}^T \|G_t - \bar{G}\|_1 = \sum_{t=1}^T \|G_t - \hat{G}\|_1 + \sum_{t=1}^T \|\hat{G} - \bar{G}\|_1 \leq \sum_{t=1}^T \|G_t - \hat{G}\|_1 + \frac{1}{T} \sum_{t=1}^T \|G_t\|_1 \leq 2B$ and finally Inequality [\(H.1c\)](#) holds by [Theorem 2](#). Finally we observe that in the stochastic reward case with probability at least $1 - \delta$:

$$\left(T \cdot \text{OPT}_{\bar{r}, \bar{G}} - \sum_{t=1}^T r_t^\top q_t \right) - \sum_{t=1}^T r_t^\top (q^* - q_t) \leq \mathcal{E}^r.$$

Thus, if the rewards are stochastic with probability at least $1 - 14\delta$ it holds:

$$R_T \leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \Lambda \mathcal{E}^\dagger + 2\Lambda B + \mathcal{E}^r, \quad V_T \leq \frac{1}{\eta} \Lambda + \mathcal{E}^\dagger$$

and if the rewards are adversarial with probability at least $1 - 13\delta$ it holds:

$$R_T \leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \Lambda \mathcal{E}^\dagger + 2\Lambda B, \quad V_T \leq \frac{1}{\eta} \Lambda + \mathcal{E}^\dagger$$

which concludes the proof. □

Theorem 7. Suppose that [Condition 2](#) does not hold and the constraints are generated adversarially and are parameterized given B . Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{\mathcal{O}}(T^{1/4}(\sqrt{T} + B)), \quad V_T \leq \tilde{\mathcal{O}}(T^{3/4}),$$

with probability at least $1 - 11\delta$ when the rewards are stochastic, and with probability at least $1 - 10\delta$ when the rewards are adversarial.

Proof. Similar to the proof of [Lemma 6](#) it holds with probability at least $1 - 10\delta$:

$$\begin{aligned} \sum_{t=1}^T r_t^\top q^* - \sum_{t=1}^T r_t^\top q^{P, \pi_t} &\leq \sum_{t=1}^T (r_t^\top q^* - \lambda_t^\top G_t^\top q^*) - \sum_{t=1}^T (r_t^\top q_t - \lambda_t^\top G_t^\top q_t) + \sum_{t=1}^T \lambda_t^\top G_t^\top (q^* - q_t) \\ &\leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^\dagger + 2\lambda_{1,T} B, \end{aligned}$$

therefore with probability at least $1 - 10\delta$ following the reasoning of [Lemma 5](#) it holds with adversarial rewards:

$$\sum_{t=1}^T r_t^\top q_t \geq T \cdot \text{OPT}_{\bar{r}, \bar{G}} - mT^{1/4} \mathcal{E}^\parallel - 2mT^{1/4} B - \mathcal{E}^D(\underline{0}) - \mathcal{E}^P,$$

and with stochastic rewards with probability at least $1 - 11\delta$:

$$\sum_{t=1}^T r_t^\top q_t \geq T \cdot \text{OPT}_{\bar{r}, \hat{G}} - mT^{1/4} \mathcal{E}^\parallel - 2mT^{1/4} B - \mathcal{E}^D(\underline{0}) - \mathcal{E}^P - \mathcal{E}^r.$$

Applying [Lemma 5](#) to bound the constraints violation concludes the proof. $\square$

Appendix I. Analysis with respect to the weaker baseline

In this section we will study the guarantees of [Algorithm 2](#) when the regret is computed with respect to a policy q^* that respect the constraints at each episode, i.e. $g_i^\top q^* \leq 0$ for all $i \in [m]$, for all $t \in [T]$.

Theorem 8.

Suppose that [Condition 2](#) holds and the constraints are generated adversarially. Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{\mathcal{O}}(\Lambda \sqrt{T}), \quad V_T \leq \tilde{\mathcal{O}}(\Lambda \sqrt{T}),$$

with probability at least $1 - 13\delta$ when the rewards are stochastic, and with probability at least $1 - 12\delta$ when the rewards are adversarial.

Proof. The violation can be bounded as in [Theorem 6](#). Also similarly to [Theorem 6](#) it holds with probability $1 - 12\delta$

$$\begin{aligned} \sum_{t=1}^T r_t^\top q^* - \sum_{t=1}^T r_t^\top q_t &\leq - \sum_{t=1}^T \ell_t^{\mathcal{L}, \top} q^* + \sum_{t=1}^T \ell_t^{\mathcal{L}, \top} q_t + \sum_{t=1}^T \lambda_t^\top G_t^\top q^* - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t \\ &\leq \mathcal{E}_T^P + \sum_{t=1}^T (\underline{0} - \lambda_t)^\top \sum_{h=0}^{H-1} G_t(x_h, a_h) + \sum_{t=1}^T \lambda_t^\top \left(\sum_{h=0}^{H-1} G_t(x_h, a_h) - G_t^\top q_t \right) \\ &\leq \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^\parallel \\ &\leq \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + \Lambda \mathcal{E}^\parallel. \end{aligned}$$

Finally with stochastic rewards with probability at least $1 - \delta$:

$$T \cdot \text{OPT}_{\bar{r}, \bar{G}} - \sum_{t=1}^T r_t^\top q^* \leq \mathcal{E}^r.$$

Therefore, with adversarial rewards it holds with probability at least $1 - 12\delta$:

$$R_T \leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \Lambda \mathcal{E}^\parallel, \quad V_T \leq \frac{1}{\eta} \Lambda + \mathcal{E}^\parallel,$$

and with stochastic rewards it holds with probability at least $1 - 13\delta$:

$$R_T \leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + \Lambda \mathcal{E}^\parallel + \mathcal{E}^r, \quad V_T \leq \frac{1}{\eta} \Lambda + \mathcal{E}^\parallel,$$

which concludes the proof. $\square$

Theorem 9.

Suppose that [Condition 2](#) does not hold and the constraints are generated adversarially. Then, for any $\delta \in (0, 1)$, [Algorithm 2](#) attains:

$$R_T \leq \tilde{\mathcal{O}}(T^{3/4}), \quad V_T \leq \tilde{\mathcal{O}}(T^{3/4}),$$

with probability at least $1 - 12\delta$ when the rewards are stochastic, and with probability at least $1 - 11\delta$ when the rewards are adversarial.

Proof. The violation can be bounded thanks to [Lemma 3](#), as in [Theorem 4](#). To bound the regret, notice that it holds with probability $1 - 9\delta$:

$$\begin{aligned} \sum_{t=1}^T r_t^\top q^* - \sum_{t=1}^T r_t^\top q_t &\leq - \sum_{t=1}^T \ell_t^{\mathcal{L}, \top} q^* + \sum_{t=1}^T \ell_t^{\mathcal{L}, \top} q_t + \sum_{t=1}^T \lambda_t^\top G_t^\top q^* - \sum_{t=1}^T \lambda_t^\top G_t^\top q_t \\ &\leq \mathcal{E}_T^P + \sum_{t=1}^T (\underline{0} - \lambda_t)^\top \sum_{h=0}^{H-1} G_t(x_h, a_h) + \sum_{t=1}^T \lambda_t^\top \left(\sum_{h=0}^{H-1} G_t(x_h, a_h) - G_t^\top q_t \right) \\ &\leq \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + \lambda_{1,T} \mathcal{E}^\parallel \\ &\leq \mathcal{E}_T^P + \mathcal{E}_T^D(\underline{0}) + mT^{1/4} \mathcal{E}^\parallel. \end{aligned}$$

Finally with stochastic rewards with probability at least $1 - \delta$:

$$T \cdot \text{OPT}_{\bar{r}, \bar{G}} - \sum_{t=1}^T r_t^\top q^* \leq \mathcal{E}^r.$$

Therefore, with adversarial rewards it holds with probability at least $1 - 11\delta$:

$$R_T \leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + mT^{1/4} \mathcal{E}^\parallel, \quad V_T \leq \frac{4T^{1/4}}{\eta},$$

and with stochastic rewards it holds with probability at least $1 - 12\delta$:

$$R_T \leq \mathcal{E}^P + \mathcal{E}^D(\underline{0}) + mT^{1/4} \mathcal{E}^\parallel + \mathcal{E}^r, \quad V_T \leq \frac{4T^{1/4}}{\eta},$$

which concludes the proof. $\square$

Appendix J. Auxiliary lemmas

Lemma 6 (Adapted from [19] Lemma C.4).

$$\eta_t \hat{Q}_t(x, a) \leq \frac{1}{2} \quad \wedge \quad \eta_t B_t(x, a) \leq \frac{1}{2H}.$$

Proof. Recall $\gamma = 2\eta_t H \Xi_t$. Thus, it holds:

$$\eta_t \hat{Q}_t(x, a) \leq \frac{\eta_t H \Xi_t}{\gamma} = \frac{\eta_t H \Xi_t}{2\eta_t H \Xi_t} = \frac{1}{2}$$

and

$$\eta_t b_t(x, a) = \frac{3\eta_t H \Xi_t \gamma + \eta_t \Xi_t H(\bar{q}_t(x, a) - \underline{q}_t(x, a))}{\bar{q}_t(x, a) + \gamma} \leq 3\eta_t \Xi_t H + \eta_t H \Xi_t = 2\gamma.$$

Finally,

$$\begin{aligned} \eta_t B_t(x, a) &\leq H \left(1 + \frac{1}{H}\right)^H \eta_t \sup_{x', a'} b_t(x', a') \\ &\leq 3H2\gamma \\ &= 6H\gamma \\ &= \frac{6H}{C\sqrt{T}} \\ &= \frac{6H}{252|X||A|H\sqrt{T}} \\ &\leq \frac{1}{42H} \\ &\leq \frac{1}{2H}. \end{aligned}$$

This concludes the poof. $\square$

Lemma 7 (Adapted from [19], Lemma B.1). *If the following inequality holds:*

$$\begin{aligned} &\sum_x q^*(x) \sum_{t=1}^{t_2} \sum_a (\pi_t(a|x) - \pi^*(a|x)) (\mathcal{Q}_t^{\pi_t}(x, a) - B_t(x, a)) \\ &\leq o(T) + \sum_{t=1}^{t_2} V^{\pi^*}(x_0; b_t) + \frac{1}{H} \sum_{t=1}^{t_2} \sum_{x,a} q^*(x) \pi_t(a|x) B_t(x, a), \end{aligned} \quad (\text{J.1})$$

with B_t defined as

$$B_t(x, a) = b_t(x, a) + \left(1 + \frac{1}{H}\right) \mathbb{E}_{x' \sim P(\cdot|x,a)} \mathbb{E}_{a' \sim \pi_t(\cdot|x')} [B_t(x', a')] \quad \forall t \in [T], \forall x \in X, \forall a \in A, \quad (\text{J.2})$$

then it holds that:

$$R_{t_1, t_2} \leq o(T) + 3 \sum_{t=t_1}^{t_2} \hat{V}^{\pi_t}(x_0; b_t).$$

Proof. The proof is analogous to the one proposed by [19], Lemma B.1, since the proof is episode based and then the sum over t is taken. $\square$

Lemma 8 (Adapted from [19], Lemma A.1). *Let $\mathcal{F}_0, \dots, \mathcal{F}_T$ be a filtration and $X_1, \dots, X_T$ be real random variables such that X_t is $\mathcal{F}_t$ -measurable, $\mathbb{E}[X_t|\mathcal{F}_t] = 0$, $|X_t| \leq b$ for all $t \in [T]$ and $\sum_{t=t_1}^{t_2} \mathbb{E}[X_t^2|\mathcal{F}_t] \leq V_{t_1, t_2}$ for some fixed $V_{t_1, t_2} > 0$ and $b > 0$ for every $t_1, t_2 \in [T]$ such that $1 \leq t_1 \leq t_2 \leq T$. Then with probability at least $1 - \delta$ it holds simultaneously for all $[t_1, \dots, t_2] \subset [T]$:*

$$\sum_{t=t_1}^{t_2} X_t \leq \frac{V_{t_1, t_2}}{b} + b \log \left(\frac{T^2}{\delta} \right).$$

Proof. For all $\delta' \in (0, 1)$ by Lemma A.1 [19] it holds:

$$\mathbb{P} \left(\sum_{t=t_1}^{t_2} X_t \geq \frac{V_{t_1, t_2}}{b} + b \log \left(\frac{1}{\delta'} \right) \right) \leq \delta'.$$

It is sufficient to consider the intersection of all events for all possible intervals $[t_1, \dots, t_2]$, that are less than T^2 .

$$\mathbb{P} \left(\bigcap_{t_1, t_2} \left\{ \sum_{t=t_1}^{t_2} X_t \geq \frac{V_{t_1, t_2}}{b} + b \log \left(\frac{1}{\delta'} \right) \right\} \right) \leq T^2 \delta'.$$

To conclude the proof we take δ as $T^2 \delta'$. $\square$

Consider a loss function $f_t(x, a) \in [0, Z]$, for all $t \in [T], (x, a) \in X \times A$, with $Z > 0$. Define another function $\tilde{f}_t \in [0, Z]^{|X \times A|}$. If we define the estimator $\hat{f}_t(x, a) = \frac{\tilde{f}_t(x, a) \mathbb{I}_t(x, a)}{\bar{q}_t(x, a) + \gamma}$ where $\mathbb{E}[\tilde{f}_t(x, a)] = f_t(x, a)$, we can state the following result.

Lemma 9 (Adapted from [18]). *For every sequence of functions $\alpha_1, \dots, \alpha_T$ such that $\alpha_t \in [0, \frac{2\gamma}{Z}]^{|X \times A|}$ is $\mathcal{F}_t$ measurable for all $t \in [T]$, we have with probability at least $1 - \delta$ that simultaneously for all $t_1, t_2 \in [T]$ such that $1 \leq t_1 \leq t_2 \leq T$ it holds:*

$$\sum_{t=t_1}^{t_2} \sum_{x, a} \alpha_t(x, a) \left(\hat{f}_t(x, a) - \frac{q_t(x, a)}{\bar{q}_t(x, a)} f_t(x, a) \right) \leq H \ln \left(\frac{HT^2}{\delta} \right).$$

Proof.

$$\begin{aligned} \hat{\ell}_t(x, a) &= \frac{\tilde{f}_t(x, a) \mathbb{I}_t(x, a)}{\bar{q}_t(x, a) + \gamma} \\ &\leq \frac{\tilde{f}_t(x, a) \mathbb{I}_t(x, a)}{\bar{q}_t(x, a) + \frac{\tilde{f}_t(x, a)}{Z} \gamma} \\ &= \frac{\mathbb{I}_t(x, a) Z}{2\gamma} \frac{2\gamma \frac{\tilde{f}_t(x, a)}{Z}}{\bar{q}_t(x, a) + \gamma \frac{\tilde{f}_t(x, a)}{Z}} \\ &= \frac{\mathbb{I}_t(x, a) Z}{2\gamma} \frac{2\gamma \frac{\tilde{f}_t(x, a)}{Z \bar{q}_t(x, a)}}{1 + \gamma \frac{\tilde{f}_t(x, a)}{Z \bar{q}_t(x, a)}} \\ &\leq \frac{Z}{2\gamma} \ln \left(1 + 2\gamma \frac{\mathbb{I}_t(x, a) \tilde{f}_t(x, a)}{Z \bar{q}_t(x, a)} \right). \end{aligned}$$

For each layer $h \in [H]$ we define $\hat{S}_{t, h} := \sum_{x \in X_h, a \in A} \alpha_t(x, a) \hat{f}_t(x, a)$ and $S_{t, h} := \sum_{x \in X_h, a \in A} \alpha_t(x, a) \frac{q_t(x, a)}{\bar{q}_t(x, a)} f_t(x, a)$.

$$\begin{aligned} \mathbb{E}_t[\exp(\hat{S}_{t, h})] &= \mathbb{E} \left[\exp \left(\sum_{x \in X_h, a \in A} \alpha_t(x, a) \hat{f}_t(x, a) \right) \right] \\ &\leq \mathbb{E} \left[\exp \left(\sum_{x \in X_h, a \in A} \alpha_t(x, a) \frac{Z}{2\gamma} \ln \left(1 + 2\gamma \frac{\mathbb{I}_t(x, a) \tilde{f}_t(x, a)}{Z \bar{q}_t(x, a)} \right) \right) \right] \\ &\leq \mathbb{E} \left[\prod_{x \in X_h, a \in A} \left(1 + \alpha_t(x, a) \frac{\mathbb{I}_t(x, a) \tilde{f}_t(x, a)}{\bar{q}_t(x, a)} \right) \right] \\ &\leq 1 + \sum_{x \in X_h, a \in A} \alpha_t(x, a) \frac{q_t(x, a) f_t(x, a)}{\bar{q}_t(x, a)} = 1 + S_{t, h} \\ &\leq \exp(S_{t, h}) \end{aligned}$$

For each interval $[t_1, \dots, t_2] \subset [T]$ it holds:

$$\mathbb{P} \left[\sum_{t=t_1}^{t_2} (\hat{S}_{t, h} - S_{t, h}) \geq \ln \left(\frac{H}{\delta'} \right) \right] \leq \frac{\delta'}{H}.$$

Taking the intersection event for all intervals $[t_1, \dots, t_2] \subset [T]$:

$$\mathbb{P} \left[\bigcap_{t_1, t_2} \left\{ \sum_{t=t_1}^{t_2} (\hat{S}_{t,h} - S_{t,h}) \geq \ln \left(\frac{H}{\delta'} \right) \right\} \right] \leq T^2 \frac{\delta'}{H}.$$

$$\delta = T^2 \delta',$$

and

$$\mathbb{P} \left[\bigcap_{t_1, t_2} \left\{ \sum_{t=t_1}^{t_2} (\hat{S}_{t,h} - S_{t,h}) \geq \ln \left(\frac{HT^2}{\delta} \right) \right\} \right] \leq \frac{\delta}{H}.$$

Finally we take the sum over $h \in [H]$:

$$\mathbb{P} \left[\sum_{t=t_1}^{t_2} \sum_{x,a} \alpha_t(x, a) \left(\hat{f}_t(x, a) - \frac{q_t(x, a)}{u_t(x, a)} f_t(x, a) \right) \leq H \ln \left(\frac{HT^2}{\delta} \right) \right] \leq \delta.$$

This concludes the proof. $\square$

Corollary 1. Given $\delta \in (0, 1)$, it holds with probability at least $1 - 2\delta$ simultaneously for all $t_1, t_2 \in [T]$ such that $1 \leq t_1 \leq t_2 \leq T$:

$$\sum_{t=t_1}^{t_2} \sum_{x,a} \left(\hat{f}_t(x, a) - f_t(x, a) \right) \leq \frac{ZH}{2\gamma} \ln \left(\frac{HT^2}{\delta} \right).$$

Lemma 10. Let $\{\pi_t\}_{t=1}^T$ policies, then for any collection of transition $P_t^x \in \mathcal{P}_{i(t)}$ with probability at least $1 - 2\delta$,

$$\sum_{t=1}^T \|q^{P, \pi_t} - q^{P_t^x, \pi_t}\|_1 \leq 2H|X|^2 \sqrt{2T \ln \left(\frac{H|X|}{\delta} \right)} + 3H|X|^2 \sqrt{2T|A| \ln \left(\frac{T|X|^2|A|}{\delta} \right)}.$$

Proof. It holds:

$$\begin{aligned} \sum_{t=1}^T \|q^{P, \pi_t} - q^{P_t^x, \pi_t}\|_1 &= \sum_{t=1}^T \sum_{x,a} |q^{P, \pi_t}(x, a) - q^{P_t^x, \pi_t}(x, a)| \\ &\leq \sum_{t=1}^T \sum_{x,a} \sum_{x'} |q^{P, \pi_t}(x', a) - q^{P_t^x, \pi_t}(x', a)| \\ &= \sum_x \sum_{t=1}^T \sum_{x',a} |q^{P, \pi_t}(x', a) - q^{P_t^x, \pi_t}(x', a)| \\ &\leq \sum_x \left(2H|X| \sqrt{2T \ln \left(\frac{H|X|}{\delta} \right)} + 3H|X| \sqrt{2T|A| \ln \left(\frac{T|X|^2|A|}{\delta} \right)} \right) \\ &\leq |X| \left(2H|X| \sqrt{2T \ln \left(\frac{|X|H}{\delta} \right)} + 3H|X| \sqrt{2T|A| \ln \left(\frac{T|X|^2|A|}{\delta} \right)} \right), \end{aligned}$$

by Lemma 13, taking the union bound over X ($\delta' = \frac{\delta}{|X|}$). This concludes the proof. $\square$

Appendix K. Auxiliary lemmas from existing works

Lemma 11 (Stradi et al. [1] Lemma D.2). For $\eta \leq \frac{1}{mH}$ and $\frac{M}{\rho} > 4$, if $\|\lambda_{t_2+1}\|_1 > \frac{2HM}{\rho^2}$ and $\|\lambda_{t_1}\|_1 \leq \frac{4H}{\rho}$ it holds:

$$(t_2 - t_1 + 1) > \frac{M}{\rho^2 m \eta}$$

Lemma 12 (Rosenberg and Mansour [16]). For any $\delta \in (0, 1)$

$$\|P(\cdot|x, a) - \bar{P}_i(\cdot|x, a)\|_1 \leq \sqrt{\frac{2|X_{h(x)+1}| \ln \left(\frac{T|X||A|}{\delta} \right)}{\max\{1, N_i(x, a)\}}},$$

simultaneously for all $(x, a) \in X \times A$ and for all epochs with probability at least $1 - \delta$.

Lemma 13 (Rosenberg and Mansour [16]). Let $\{\pi_t\}_{t=1}^T$ policies and let $\{P_t\}_{t=1}^T$ transition functions such that $q^{P_t, \pi_t} \in \Delta(\mathcal{P}_{i(t)})$ for every $t \in [T]$. Then with probability at least $1 - 2\delta$,

$$\sum_{t=1}^T \|q^{P, \pi_t} - q^{P_t, \pi_t}\|_1 \leq 2H|X| \sqrt{2T \ln \left(\frac{H}{\delta} \right)} + 3H|X| \sqrt{2T|A| \ln \left(\frac{T|X||A|}{\delta} \right)}.$$

Appendix L. Numerical experiments

In this section, we provide a numerical validation of the behavior of PDB-PS in simple CMDP instances with bandit feedback. The goal of these experiments is to illustrate the two main phenomena emphasized by the theory: our approach keeps the cumulative constraint violations and the regret controlled, and it avoids the computational cost of occupancy-measure updates.

Experimental setup. We first consider CMDPs with a single decision state, so that the occupancy measure induced by a policy coincides with a distribution over actions. Thus, for a policy $\pi_t \in \Delta(\mathcal{A})$, the regret and violation are:

$$R_T = \sum_{t=1}^T r_t^\top \pi^* - \sum_{t=1}^T r_t^\top \pi_t, \quad V_T = \max_{i \in [m]} \sum_{t=1}^T G_{t,i}^\top \pi_t,$$

where π^* is the best feasible policy. We compare PDB-PS against a simple random greedy baseline that samples an action uniformly at random at each episode.

We report results on five CMDP instances: three stochastic instances with $|\mathcal{A}| \in \{4, 8, 16\}$ actions and two adversarial instances with switching or cyclic rewards. In stochastic instances, the observed bandit feedback is perturbed by independent bounded noise, while regret and violation are evaluated on the underlying mean instance. In adversarial instances, the sequences $\{r_t\}_{t=1}^T$ and $\{G_t\}_{t=1}^T$ are deterministic non-stationary sequences. Each curve is averaged over 30 independent runs. Solid curves denote empirical means, while the shaded regions and dotted envelopes show 95% confidence intervals.

Overall, Figs. L.1–L.5 show that PDB-PS consistently controls the cumulative constraint violation across stochastic and adversarial feedback regimes. The random greedy baseline can sometimes obtain competitive regret by sampling infeasible high-reward actions,

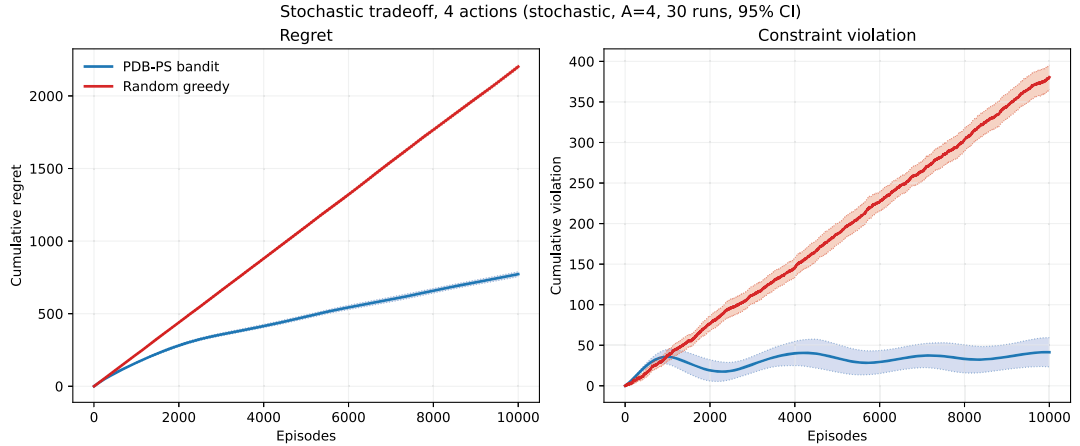


Fig. L.1. Single-state stochastic CMDP with $|\mathcal{A}| = 4$ actions. PDB-PS achieves substantially smaller regret than the random greedy baseline, while also keeping the cumulative violation much smaller.

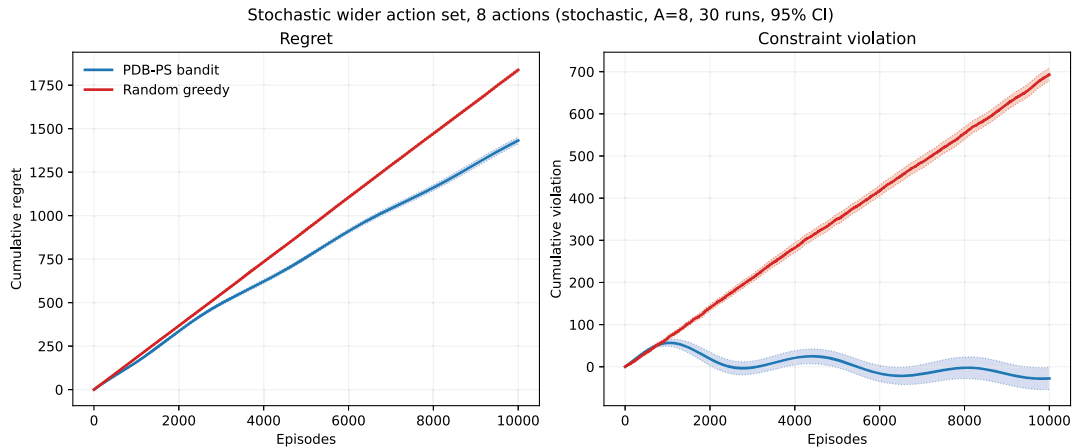


Fig. L.2. Single-state stochastic CMDP with $|\mathcal{A}| = 8$ actions. The random greedy baseline incurs large violation because it keeps sampling high-reward infeasible actions, whereas PDB-PS maintains a much lower violation.



Fig. L.3. Adversarial switching-reward CMDP with $|A| = 4$ actions. PDB-PS tracks the changing reward structure and obtains lower regret than the random greedy baseline. In this instance, the random baseline happens to remain nearly feasible, but it pays for this with larger regret.

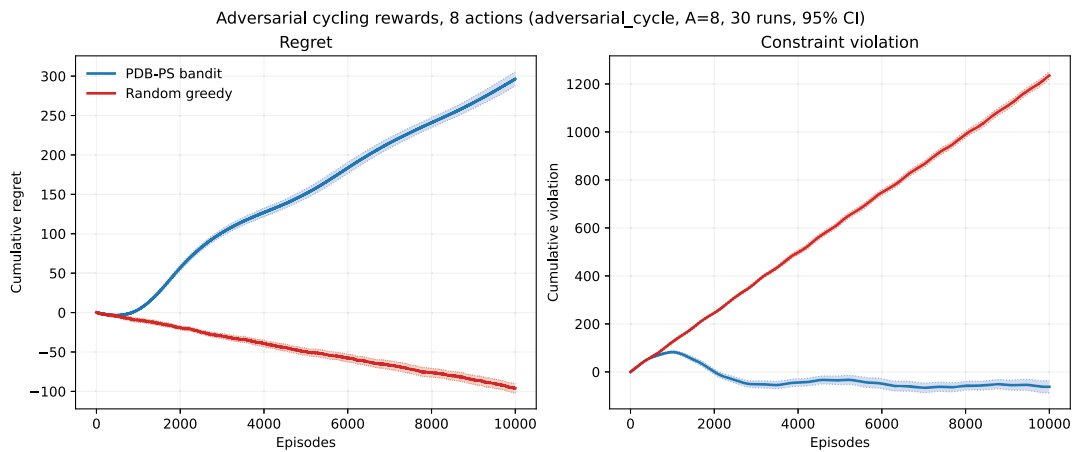


Fig. L.4. Adversarial cyclic-reward CMDP with $|A| = 8$ actions. The random greedy baseline can obtain low, and even negative, regret because it frequently samples high-reward actions that violate the constraints. This comes at the price of linear violation.

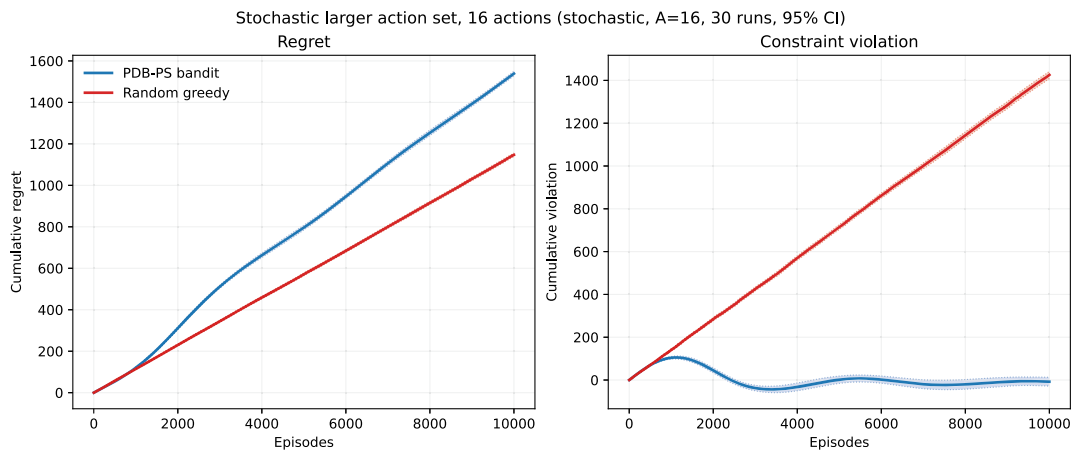


Fig. L.5. Single-state stochastic CMDP with $|A| = 16$ actions. As the action set grows, uniformly random exploration becomes increasingly costly in terms of constraint violation.

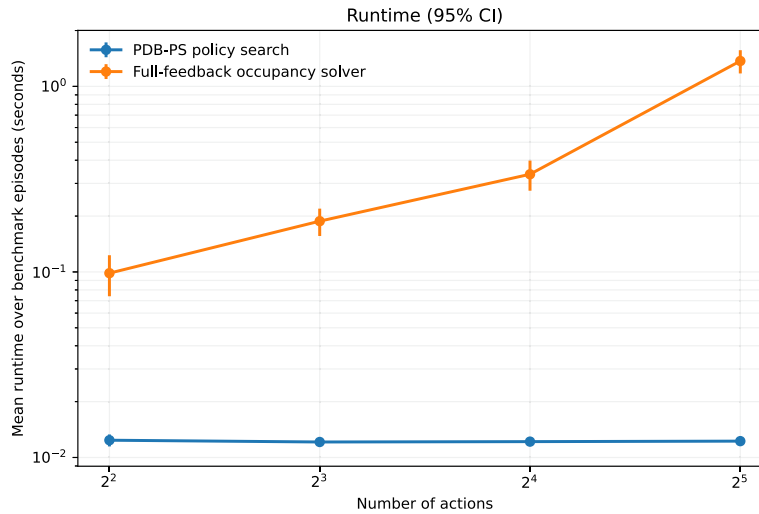


Fig. L.6. Runtime comparison between the policy-search update of PDB-PS and [1]. The gap grows quickly with the number of actions, highlighting the computational advantage of avoiding repeated optimization over the occupancy-measure polytope.

as in Figs. L.4 and L.5, but this behavior produces large cumulative violations. This confirms that regret alone is not an informative performance measure in CMDPs and that the joint behavior of R_T and V_T is essential.

Runtime comparison. We also compare the runtime of the policy-search update used by PDB-PS with a full-feedback occupancy-measure update in the spirit of [1]. The latter update solves, at each episode, an entropy-regularized occupancy-measure optimization problem over the CMDP occupancy measure polytope using a generic constrained solver, while PDB-PS performs state-wise multiplicative-weights updates. For this comparison we use a small episodic CMDP with more than one decision state, $m = 8$ constraints, and vary the number of actions. Runtime is averaged over 4 independent repeats; error bars are 95% confidence intervals.

Fig. L.6 shows that the occupancy-measure solver becomes rapidly more expensive as the action set grows, whereas the policy-search update has an almost flat runtime profile in this range. This supports the algorithmic motivation for using policy optimization rather than full-feedback occupancy-measure optimization.

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